

CAPSULES





INTRODUCTION

- The word capsule is derived from the Latin *capsula*, meaning a small box.
- In pharmacy the word is used to describe an edible package made from gelatin or other suitable material which is filled with medicines to produce a unit dosage, mainly for oral use.

Overview of capsule

- Capsules are solid dosage forms in which the drug or inert substances are enclosed within a hard or soft shell.
- The shell is normally made from gelatin and results in a simple, easy - to - swallow formulation with no requirement for a further coating step.

- HGCs are used in most commercial medicated capsules.
- The empty capsule shells are made of gelatin, water and plasticizer and may be colored with various FD and C and D & C dyes.
- Capsules are made opaque by adding agents such as titanium dioxide.

Advantage of capsule over liquid dosage form

- No need for spoon or other measuring devices
- Tasteless when swallowed
- Readily identified by color, printed name
- Available in variety of strengths
- Lower manufacturing and packaging cost
- More stable and has longer shelf life
- Designed to provide modified drug release

Advantage of capsule **over tablets**

- Elegant and attractive in appearance
- Easy to swallow than tablets
- Shell can be opacified to protect drug from light
- Mask unpleasant taste and odour of drug
- Provide ready availability of the contained drug

Disadvantage of capsules

- Capsules are not usually used for administration of extremely soluble materials such as potassium chloride
- Capsules should not be used for highly efflorescent or deliquescent materials.

RAW MATERIALS

- The raw materials used in the manufacture of both types of capsules are similar.
- Capsule shell is composed of
 - Gelatin *
 - Plasticizers *
 - Coloring agents
 - Water *
 - Opaquing agents
 - Preservatives
 - Sugar
 - Flavoring agents
 - Medicament
 - Lubricant, ...

RAW MATERIALS

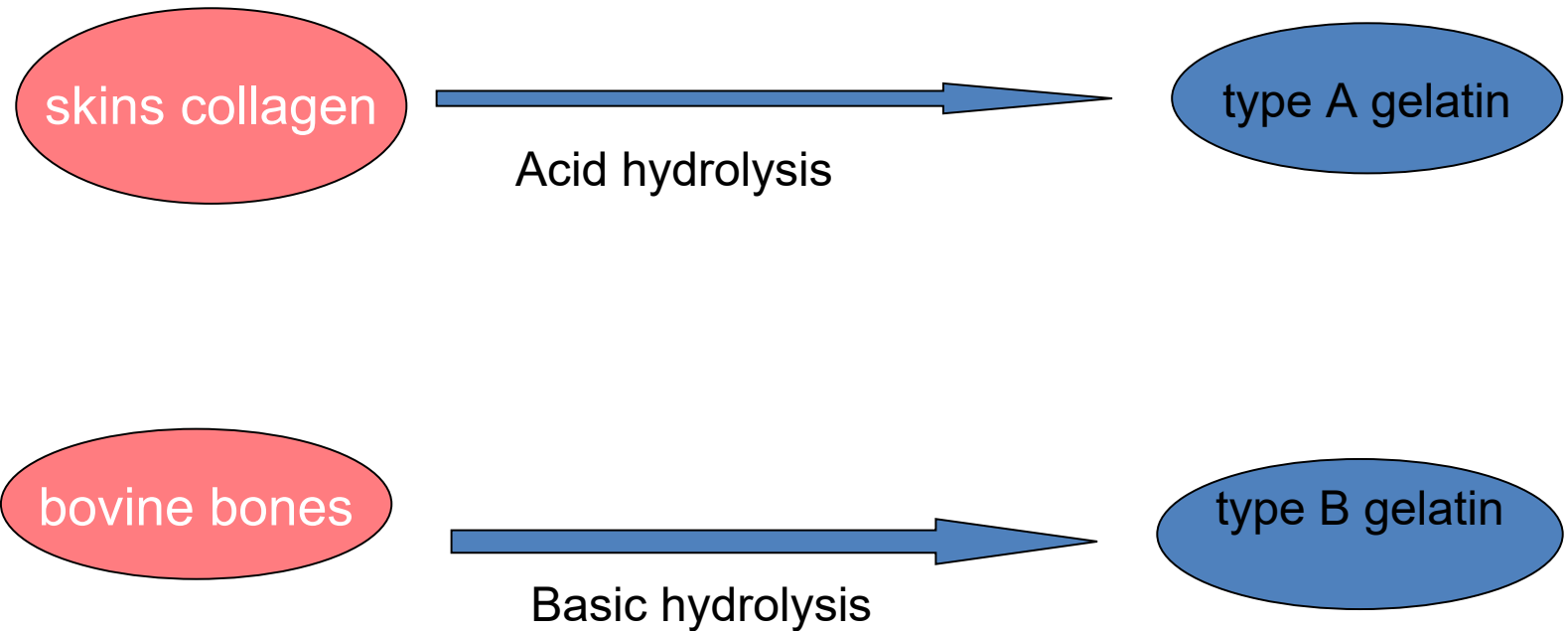
Gelatin and Alternative Materials

■ Gelatin

- ✓ is a material derived from collagen, a natural protein, which is:
 - ✓ is a fibrous material that occurs in the skin, bones, and connective tissues of animals .
- ✓ It is insoluble in water and is solubilized by hydrolysis.
- ✓ are obtained primarily from bovine bones or porcine skins.

RAW MATERIALS

- **Gelatin**



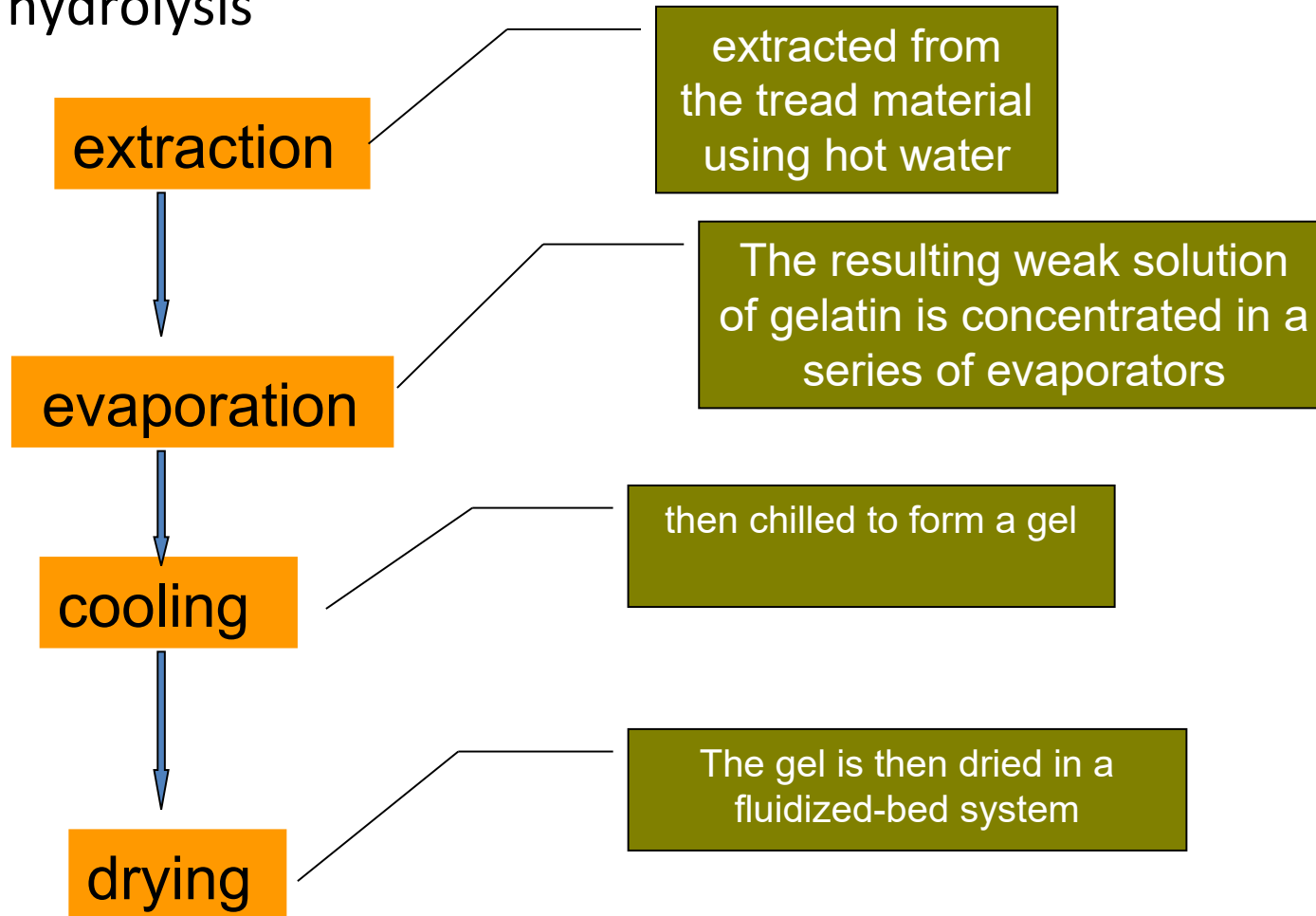
🌸 Blends of bone and pork skin gelatins of relative high gel strength are normally used for hard capsules production.

🌸 The bone gelatin produces a tough, firm film, but tends to be hazy and brittle.

🌸 The pork skin gelatin contributes plasticity and clarity to the blend, thereby reducing haze or cloudiness in the finished capsule.

RAW MATERIALS

After hydrolysis



RAW MATERIALS

- Gelatin was the first material used for the manufacture of capsules because of its unique properties.
- It is ideal because:
 - it is edible,
 - soluble at body temperature,
 - forms strong thin films,
 - undergoes a gelation process at temperatures just above ambient

- They can be either hard or soft depending on the nature of the capsule shell, with soft capsules possessing a flexible, plasticized gelatin film.
- Hard gelatin capsules are usually rigid two - piece capsules that are manufactured in one procedure and packed in another totally separate operation, whereas the formulation of soft gelatin capsules is more complex but all steps are integrated.
- Hard gelatin capsules are second only to tablets in importance as a unit dosage form.

- There is a growing interest in using non - animal - derived products for formulation of the capsule shells to address cultural, religious, and dietary requirements.

- Gelatin is obtained by the partial hydrolysis of collagen from the skin, white connective tissue and bone of animals.
- In commerce gelatin is available in the form of a fine powder or coarse powder.
- Gelatin is stable in air when dry but is subject to microbial decomposition when it becomes moist.

- Normally HGCs contain 12-16% of moisture; however, if stored in high humid environment, they become distorted and lose their rigid shape because they absorb additional moisture and if stored in an environment of extreme dryness they become brittle because some of the moisture normally present in gelatin capsule is lost.
- Therefore, it is desirable to maintain HGCs in an environment free from excessive humidity or dryness.

- Gelatin is insoluble in cold water but it does soften through absorption of up to 10 times its weight of water.
- Gelatin is soluble in hot water and in warm gastric fluid.
- In gastric gelatin capsule rapidly dissolve and expose its contents.
- Gelatin is a protein, therefore it is digested by proteolytic and then absorbed.

RAW MATERIALS

- Limitation of gelatin

- there are certain religious or dietary restrictions on its use.
- Bovine Spongiform Encephalopathy
- Water content

- Gelatin Alternative Materials

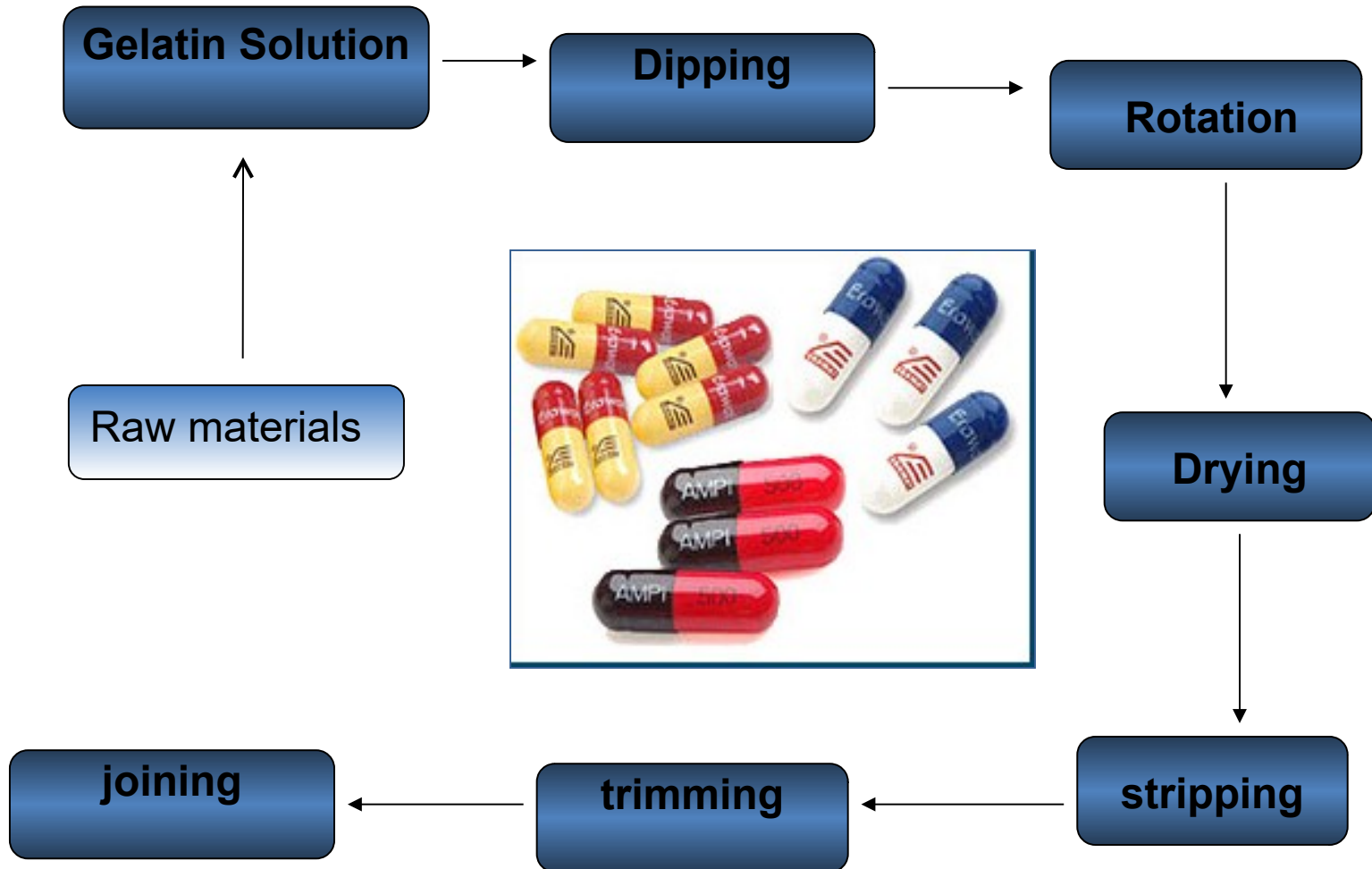
- ✓ **hydroxypropyl methylcellulose (HPMC)**
 - moisture content is much lower

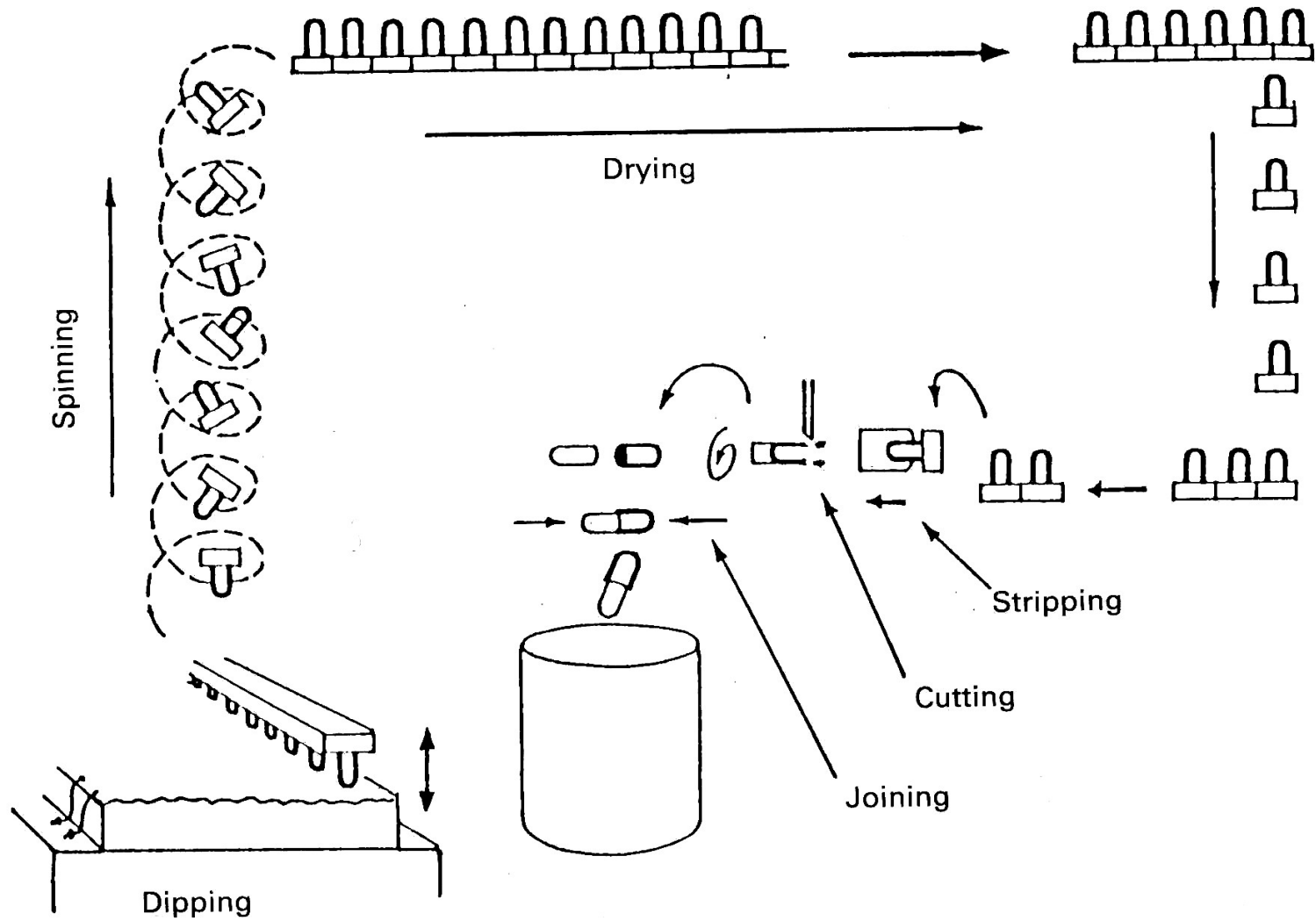
Starch capsules: A hard gelatin capsule alternative

- Starches are known to have the potential to replace the conventional capsule manufacture.
- High amylose starches have good film forming property (major property that a capsule must have) and mechanical strength.
- Starch capsules are produced by conventional dip-moulding process

- The shell is produced industrially by the mechanism for automatically/steps of capsule shell production
 - Dipping
 - Spinning
 - Drying
 - Stripping
 - Trimming/Cutting
 - Joining

CAPSULE SHELL MANUFACTURE...





Dipping

- Stainless steel mold pin controls some of the final critical dimensions of the capsule. These pins are dipped into a gelatin solution of carefully controlled viscosity to form caps and bodies simultaneously.

Spinning

- The pins are rotated to distribute the gelatin solution uniformly

Drying

- The pins are moved through a series of controlled air drying kilns for the gradual and precisely controlled removal of water.

Stripping and Trimming

- The capsules are stripped from the pin by bronze jaws and trimmed to length by stationary knives

Joining

- After being trimmed to exact length, the cap and body section are joined and ejected from the machine.
- Note: The entire cycle of the machine lasts approximately 45 minutes.

Imprinting

- Finally capsules may be imprinted with names, codes or other meanings making the product identifiable and distinguishable from other product.
- Thickness of the capsule wall is controlled by the viscosity of the gelatin solution and the speed and time of dipping. Others critical to final dimension are mold pin dimensions, precise drying, and machine controlling relating to cut length.

- In process controls include periodic monitoring and adjustment when required, film thickness, cut length of cap and body, color, moisture content.
- Inspection processes to remove imperfect capsules is performed by electronic sorting mechanism. By which having detectable visual imperfections are automatically rejected.

CAPSULE SHELL MANUFACTURE...

1. Once raw materials have been received and released by Quality Control, the gelatin and hot demineralized water are mixed under vacuum in **Stainless Steel Gelatin Melting System**.



CAPSULE SHELL MANUFACTURE...

2. After aging in stainless steel receiving tanks, the gelatin solution is transferred to **stainless steel feed tanks**



CAPSULE SHELL MANUFACTURE...

- Dyes, opacifants, and any needed water are added to the gelatin in the feed tanks to complete the gelatin preparation procedure.
- The feed tanks are then used to gravity-feed gelatin into the Capsule Machine



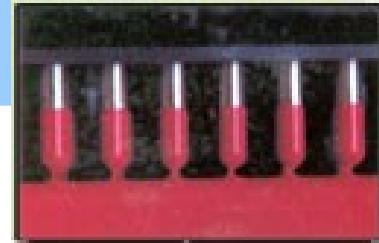
Control parameter

- Viscosity of gelatin solution

CAPSULE SHELL MANUFACTURE...

Gelatin Solution

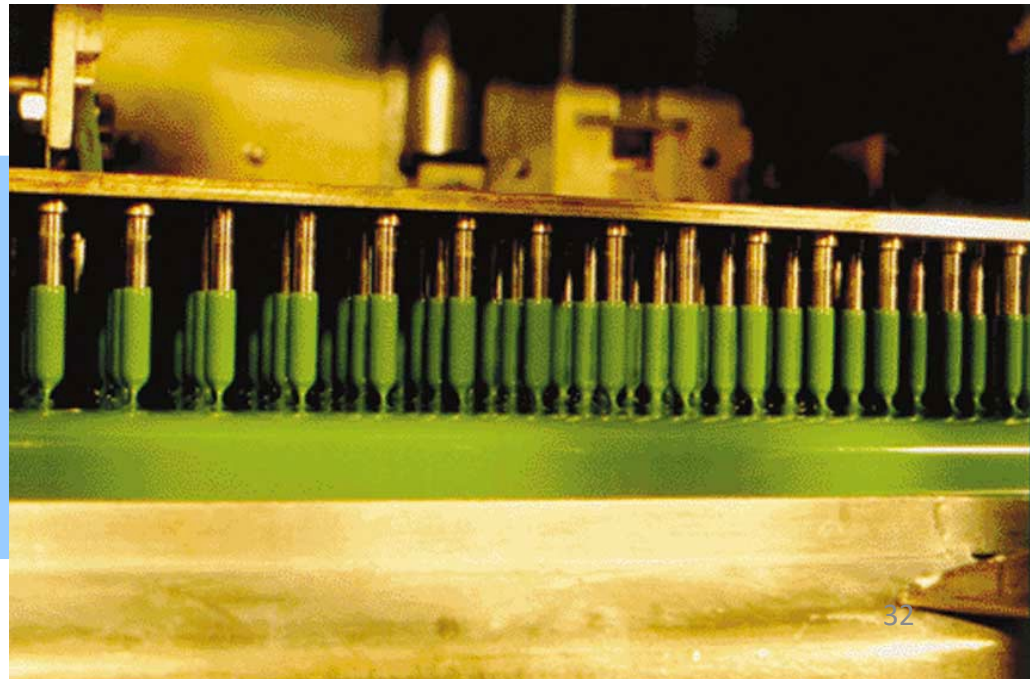
Raw gelatin and water are mixed in the ratio of 1 : 2 and processed at precise temperature



Dipping

Pairs of stainless steel pins are dipped into the dipping solution to simultaneously form the caps and bodies.

- **Control parameter**
 - Speed and time of dipping



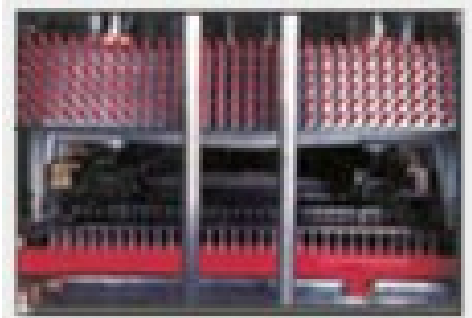
- ❖ From the feed tank, the gelatin is gravity fed to specially engineered Dipper section.
- ❖ The capsules are moulded onto stainless steel Pin Bars which are dipped into the gelatin solution.



CAPSULE SHELL MANUFACTURE...

Rotation

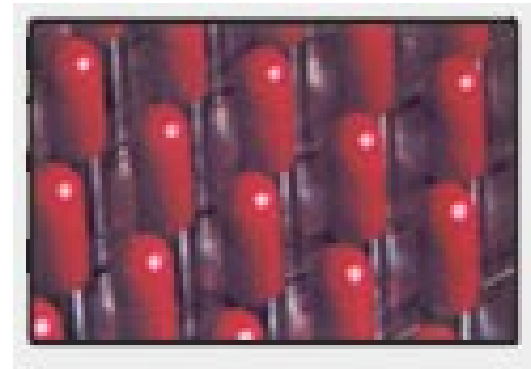
While raising, they are rotated for required distribution of gelatin on the pins.

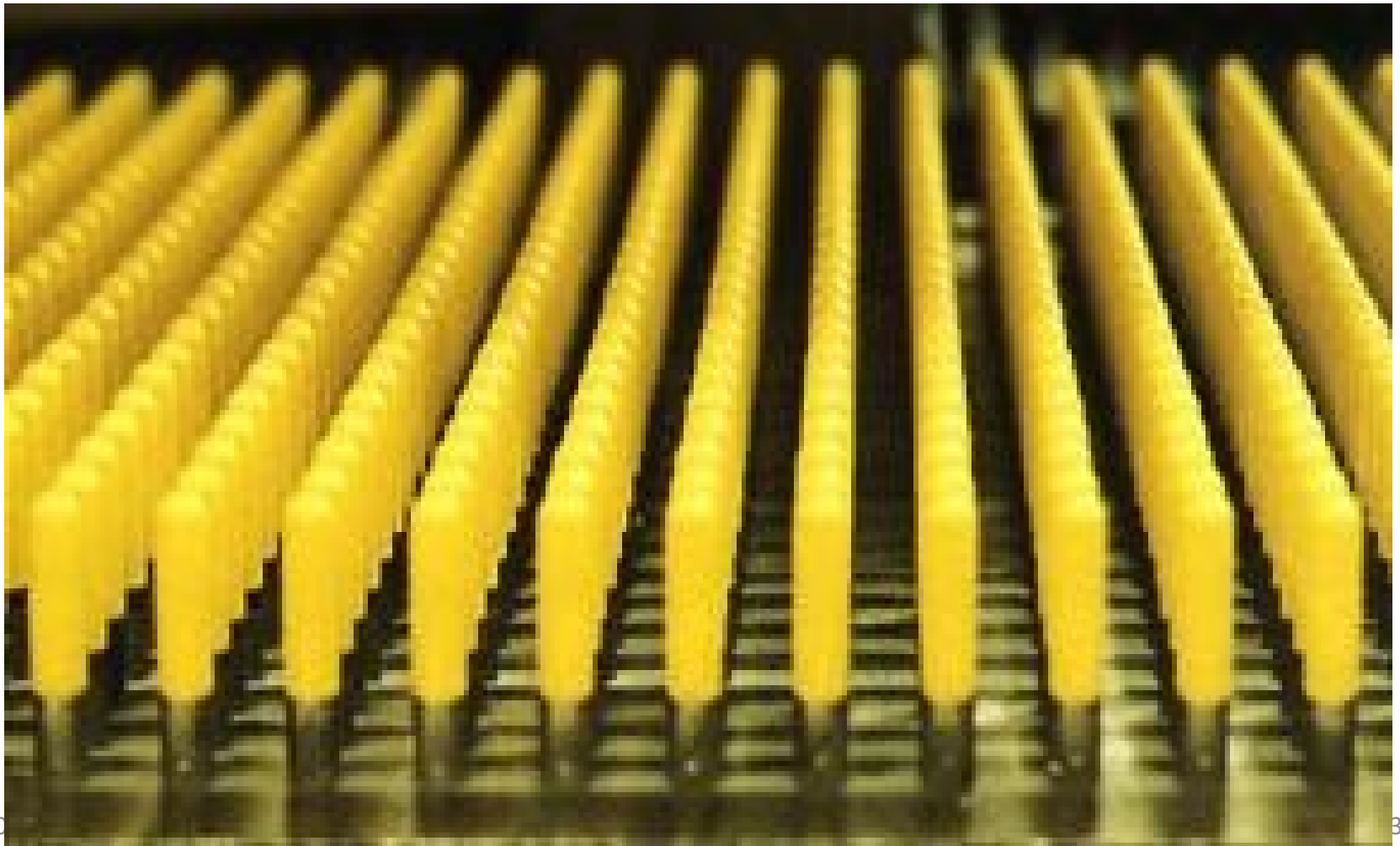


Drying

Drying is done mainly by dehumidification by passing large volumes of dry air over the pins. Only a temperature elevation a few degrees is permissible to prevent film melting

Control parameters : volume, temperature, and humidity of dry air

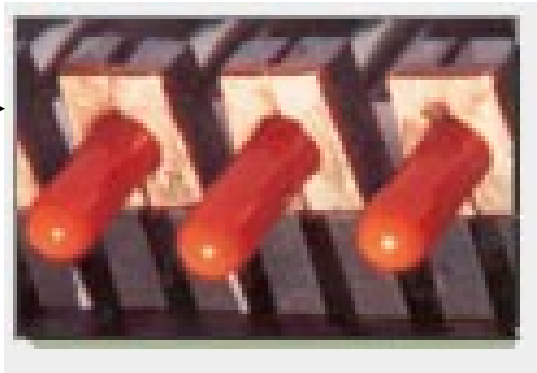




CAPSULE SHELL MANUFACTURE...

stripping

strippers remove the dried halves of capsules from the pins and place them into the collets for further operations



trimming

Caps and body halves are trimmed to narrow tolerance as per standards. The cutting are vacuumed away and collected in separate drums

CAPSULE SHELL MANUFACTURE...

joining

The capsule halves move to the centre of the machine and enter the joiner blocks. Joined capsules are then pushed on to the conveyor belt



CAPSULE SHELL MANUFACTURE...

Electronic sorting mechanism

- Inspection processes to remove imperfect capsules which were previously done visually, have recently been automated following the development and patenting of a practical electronic sorting mechanism by Eli Lilly and Company.
- This equipment mechanically orients the capsules and transports them past a series of optical scanners, at which time those having detectable visual imperfections are automatically rejected.

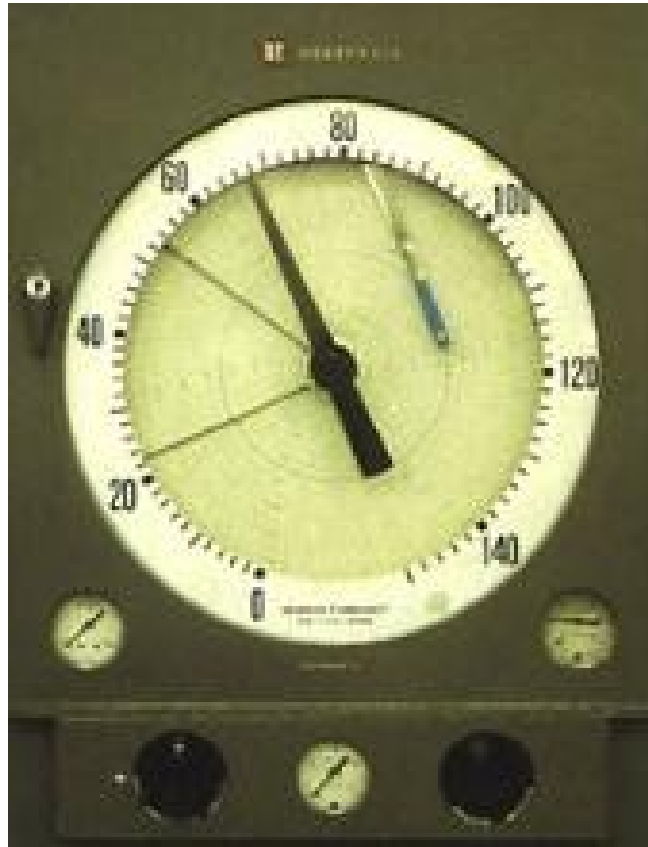
CAPSULE SHELL MANUFACTURE...

Computer Aided Systems

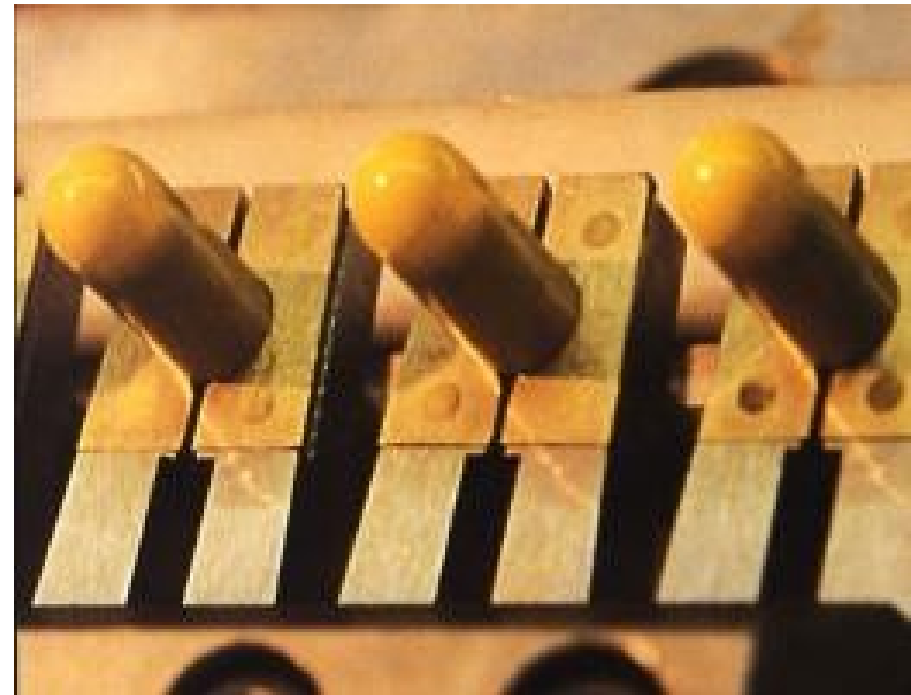
A custom made computer controlled process system monitors and controls total capsule manufacturing to ensure optimum output of highest quality capsules.



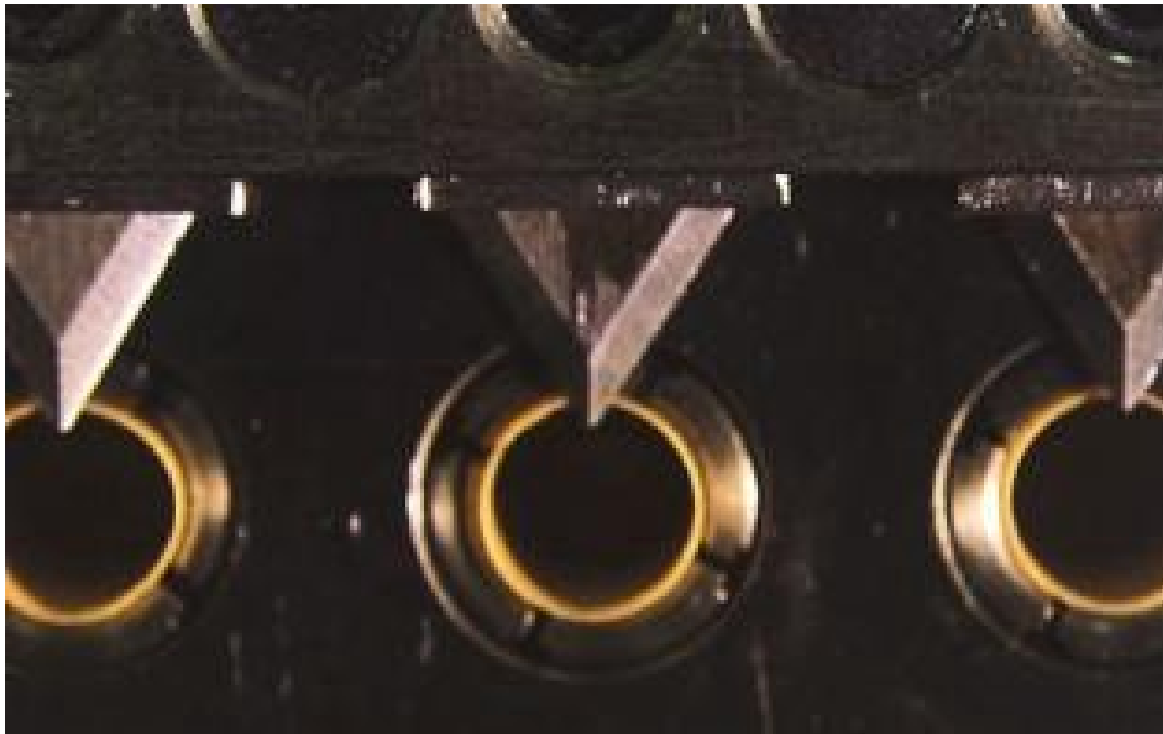
- ❖ Precision controls constantly monitor humidity, temperature, and gelatin viscosity throughout the production process.



- ❖ The Pin Bars enter the Table section which positions the capsule halves for stripping from the Pins in the Automatic section



- ❖ The cap and body lengths are precisely trimmed to the correct length to a ± 0.15 mm tolerance



- ❖ The capsule bodies and caps are joined automatically in the pre-closed position in the joiner blocks



- ❖ Finished capsules are pushed onto a conveyer belt which carries them out to a container



- ❖ Capsule quality is monitored throughout the production process including size, moisture content, single wall thickness, and colour.



- Precise bar rotation, gelatin viscosity and dipping rate all contribute to correct gelatin distribution, resulting in a homogeneous capsule wall with an exact, specified thickness.

❖ Capsules are sorted and visually inspected on specially designed Inspection Stations.

Sorting Tables

visual inspection



- ❖ Perfect prelocked capsules are imprinted with useful information like the product name, manufacturer's name or logo, and dosage details on high-speed capsule printing machines.



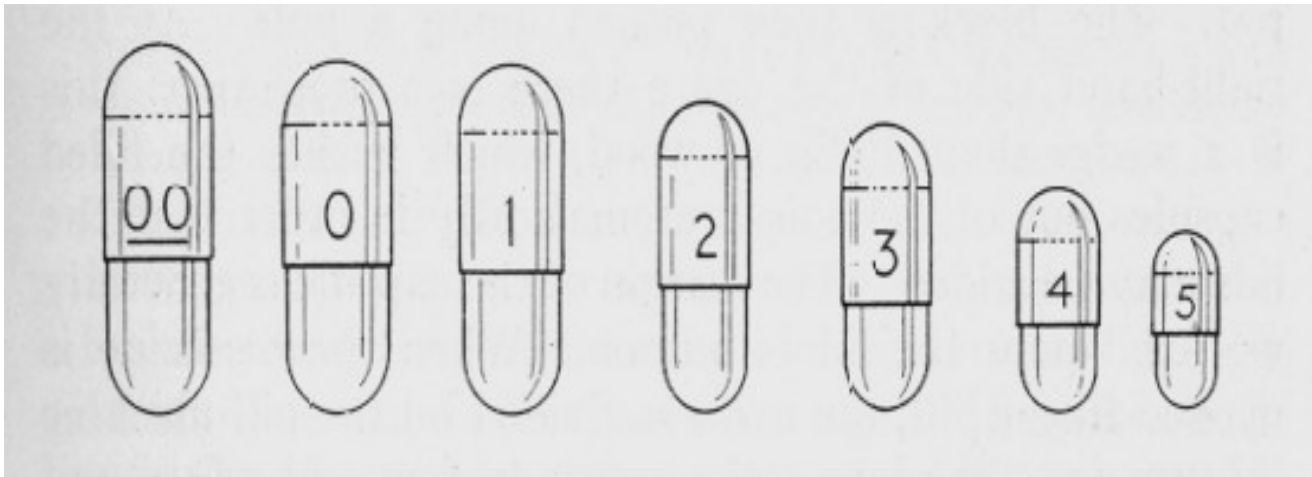
Capsule size

- Empty HGCs are manufactured in various length, diameter and capacities. The size selected for use is determined by the amount of fill material to be encapsulated.
- The density and compressibility of the fill will largely determine to what extent it may be pressed in the capsule shell.
- The size of empty of HGCs ranges from capsule number 000 (largest) to capsule number 5 (smallest).

CAPSULE FILLING

- Capsule fill capacity

Size	Volume (mL)	Fill weight ^a (g)
000	1.37	1.096
00	0.95	0.760
0	0.68	0.544
1	0.50	0.400
2	0.37	0.296
3	0.30	0.240
4	0.21	0.168
5	0.13	0.104



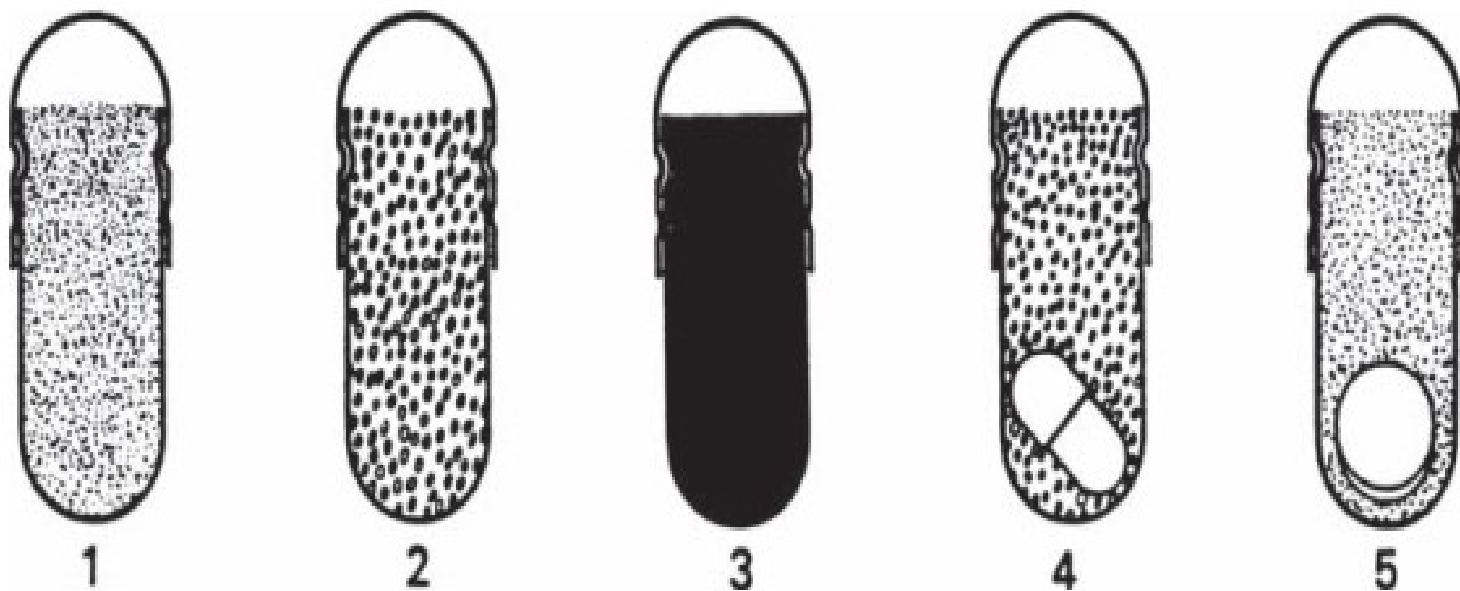
TYPES OF MATERIAL FOR FILLING INTO HARD GELATIN CAPSULES

Dry solids	Semisolids	Liquids
Powder Pellets Granules Tablets	Thermosoftening mixtures Thixotropic mixtures Pastes	Oily liquids Non-aqueous liquids

- The filling material must be compatible with the gelatin shell .
 - ✓ deliquescent or hygroscopic materials cannot be used.
 - ✓ they cannot be used for moisture - sensitive drugs.
 - ✓ All ingredients need to be free of even trace amounts of formaldehyde to minimize cross - linking of gelatin.
 - ✓ extremely soluble materials such as KCl, KBr, or NH_4Cl

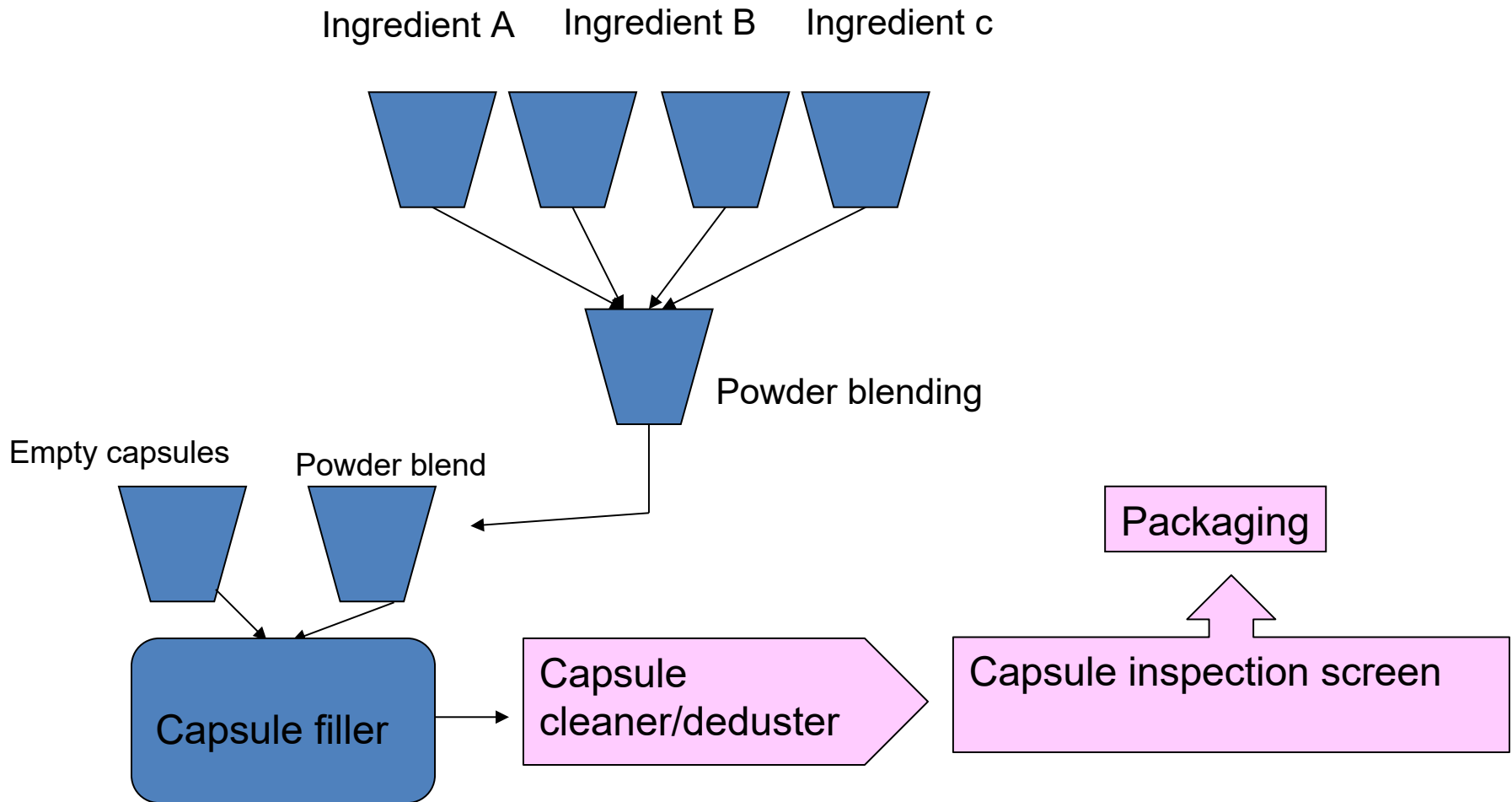
Filling operations of hard gelatin capsules

- HGCs can be filled with dry powders, pellets, small tablets, small capsules, granules,... Either in the large scale (industrial) or small scale (pharmacy) preparation of filled HGCs is divided into the following general steps.
 - Step -1: Developing and preparing the formulation and selecting the capsule size
 - Step -2: Filling the capsule shell
 - Step -3: Capsule sealing
 - Step -4: Cleaning and polishing the filled capsules



Examples of fill in hard gelatin capsules. 1, powder or granulate; 2, pellet mixture; 3, paste; 4, capsule; and 5, tablet.

Capsule shell filling: Powder filling



Step-1: Developing and preparing the formulation and selecting the capsule size

- In developing a capsule formulation, the goal is to prepare capsule with
 - Accurate dosage
 - Good bioavailability
 - Ease of filling and production
 - Stability
 - Elegance

Formulation

□ Powder formulation

- The majority of products for filling into capsules are formulated as powders.
- Among main factors in powder formulation:
 - ✓ Good flow (using free-flowing diluent and glidant)
 - ✓ No adhesion (using lubricant)

Types of excipient used in powder-filled capsules

■ Diluents (or fillers)

- The most common capsule diluents are starch and lactose.
- Inorganic salts appearing in capsule formulations include, among others, magnesium and calcium carbonate, and calcium phosphate.
- Formulations intended to be run on dosator machines may sometimes benefit from the greater compactability of diluents.

Types of excipient used in powder-filled capsules.....

❑ Glidants

- Glidants are finely dry powders added to formulations in small quantities to improve their flow characteristics.
- Glidants are thought to enhance fluidity of the formulation by one or several of possible mechanisms:
 - ✓ Reducing surface roughness by filling irregularities

Powder formulation.....

- ✓ Reducing attractive forces by physically separating the bulk powder particles;
- ✓ Acting as moisture scavengers; and
- Glidants include **colloidal silicone dioxide, cornstarch**, and **talc**.

Powder formulation.....

❑ Lubricants

- Lubricants ease the ejection of “plug”, reduce filming on pistons, and adhesion of powder to metal surfaces.
- The most effective lubricants are the hydrophobic stearates, such as **magnesium stearate, calcium stearate, and stearic acid.**

Powder formulation.....

❑ Disintegrants

- Disintegrants facilitate disruption of the powder plug when in contact with GIT fluid.

❑ Surfactants

- Surfactants may be included in capsule formulation to increase the wetting of powder mass and enhance drug dissolution.

Powder formulation.....

- The “water proofing” effect of hydrophobic lubricants may be offset by the use of surfactants.
- Common surfactants employed in capsule formulation are sodium lauryl sulfate and sodium docusate.

Formulation for position release

- Capsule formulations can be manipulated to release their contents at various positions along the gastrointestinal tract.
- Some drugs are destroyed at acid pHs, so
- Enteric product can be made:
 - either coating the filled capsule with an enteric film in a similar manner to a tablet,
 - or by formulating the contents as pellets and coating them with an enteric polymer.

- 
- The powder mix or granule must be free flowing to allow steady passage of the capsule fill from the hopper through the encapsulating equipment and into the capsule shells.

Powder handling

- The problems encountered in handling powders during mixing and filling operations are so diverse.
- Among general problems, the following are the two major ones

i- De-mixing and flow problem

- After the powder ingredients have been homogenously blended by any suitable technique, the flow of the resultant mixture must be adequate to ensure delivery of sufficient powder to the capsules at the time of filling.
- De-mixing must not occur during the powder handling in the filling equipment itself.

ii- Incompatibility between ingredients

- Physical/chemical incompatibilities between active ingredients, between diluents, or between active ingredients and/or diluents and the capsule shell may create problem.
- Because of the wide range of materials encapsulated no attempt can be made to outline specific criteria for the choice of suitable diluents.

The following are some general considerations

- Provide flow characteristics required by the equipment
- Compatibility with other ingredient and capsule shell should be studied.
- Should be approved by authority (e.g. FDA, FMHACA in Ethiopia)

Step-2: Filling the capsule shell

- HGCs are filled by one of the following methods
 - The punch method
 - Hand operating filling machine
 - Automatic industrial machine

A) The punch method

- This method is used when filling a small number of capsules in the pharmacy. It involves the following steps:
 - The pharmacist takes the precise number of empty capsules to be filled from the stock container.
 - The powder to be encapsulated is placed on a sheet of clean paper or on a glass or porcelain plate.
 - Using the spatula the powder mix is formed into a cake having the depth of $\frac{1}{4}^{\text{th}}$ to $\frac{1}{3}^{\text{rd}}$ of the capsule length.
 - Then an empty capsule body is held between the thumb and forefinger and punched vertically into the powder cake repeatedly until filled.

- The cap is replaced to close the capsule.
- Each capsule is weighed using an empty capsule as counter weight.
- Powder is added or removed until the correct weight has been placed in the capsule.
- The filled capsule is tapped so that no air spaces are visible within the content.

B- Hand operating filling machine

- Having capacities ranging from 24 to 300 caps. When efficiently operated are capsule of producing about 200 to 2000 capsules per hours

C – Automatic industrial machine

- Machines developed for industrial use automatically separate cap from body, fill body, seal capsules as desired, clean the outside of the filled capsules.
- Fill up to 165,000 capsules per hour automatically.
- The filled body must contain the accurate drug dosage, this is verified through the use of automated in process sampling and analysis.

Capsule shell filling : Powder filling

Rectification



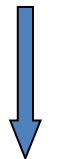
capsule separation



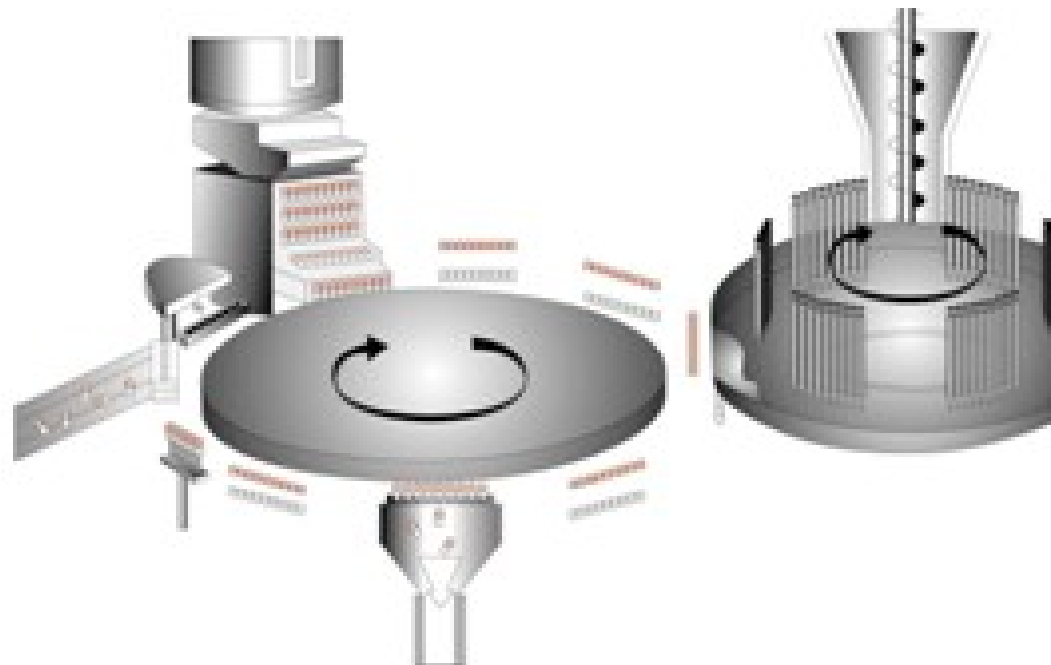
filling



closure of the capsule

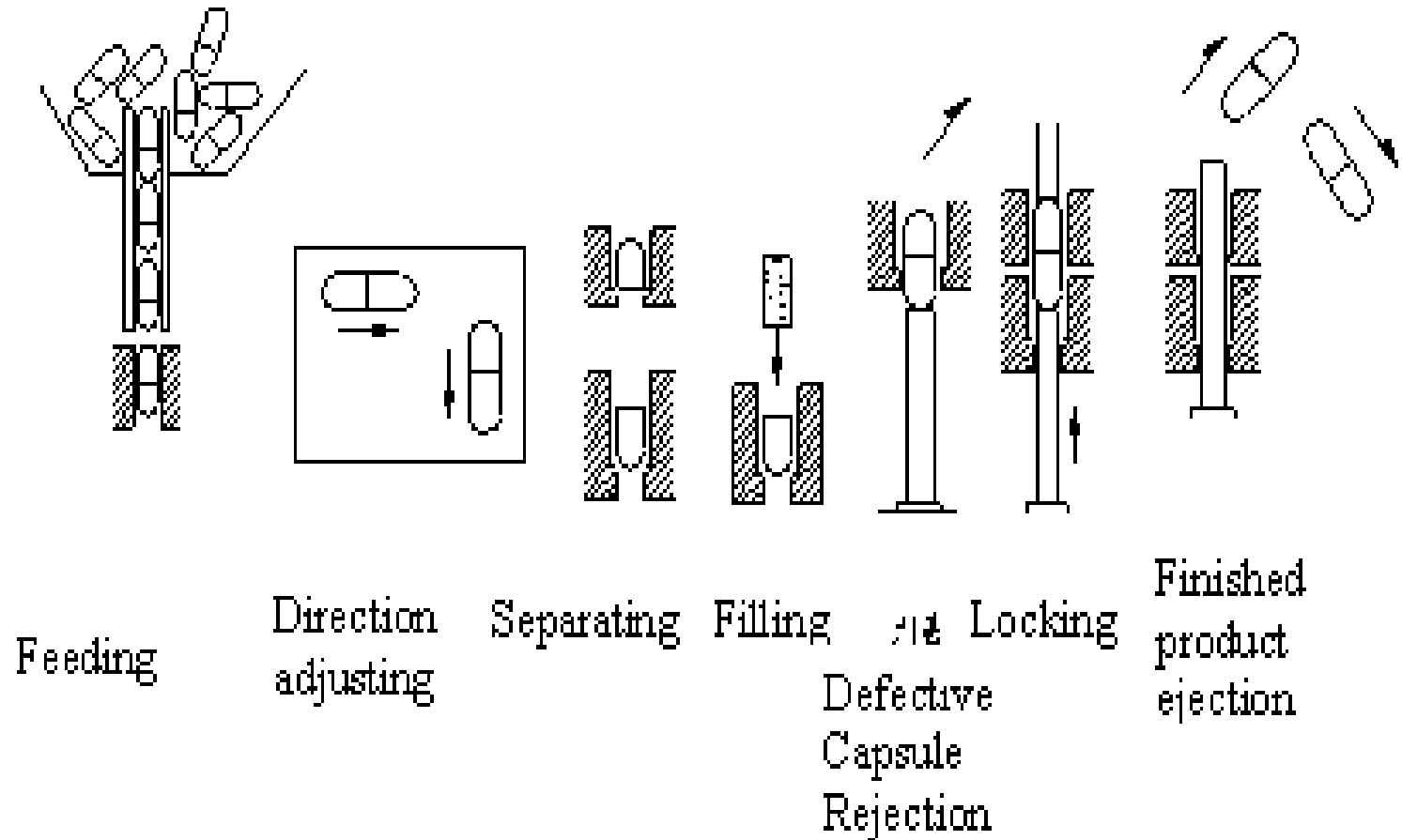


ejection of the filled and closed capsule from the machine



Capsule shell filling : Powder filling

Flow
diagram
for
Gelatin
Capsule



Capsule – filling machines

- **Filling machines** are differentiated by:
 - the way they measure the dose of material into the capsule body.
 - range in capacity from bench - top to high - output, industrial, from manual to fully automated machines.
- Those that rely on the **volume** of the shell are known as **capsule dependent**, whereas **capsule - independent** forms measure the quantity to be filled in a **separate operation**.

Capsule – filling machines...

Industrial-scale filling

- Come in great variety of shapes and sizes
- These machines may be either semi-automatic or fully automatic.

1. Semi-Automatic Machines

- require an operator to be in attendance at all times.
- depending on the skill of the operator, the formulation, and the size of the capsule being filled, these machines are capable of filling as many as 120,000 - 160,000 capsules in an 8-hour shift.

Capsule – filling machines...

Semi-Automatic Machines...

- The amount of formulation delivered to the capsule bodies depends on the **dwel time** of the bodies under the foot of the hopper.
- Fill weight uniformity depends primarily on **uniformity of flow from the hopper**.

Capsule – filling machines...

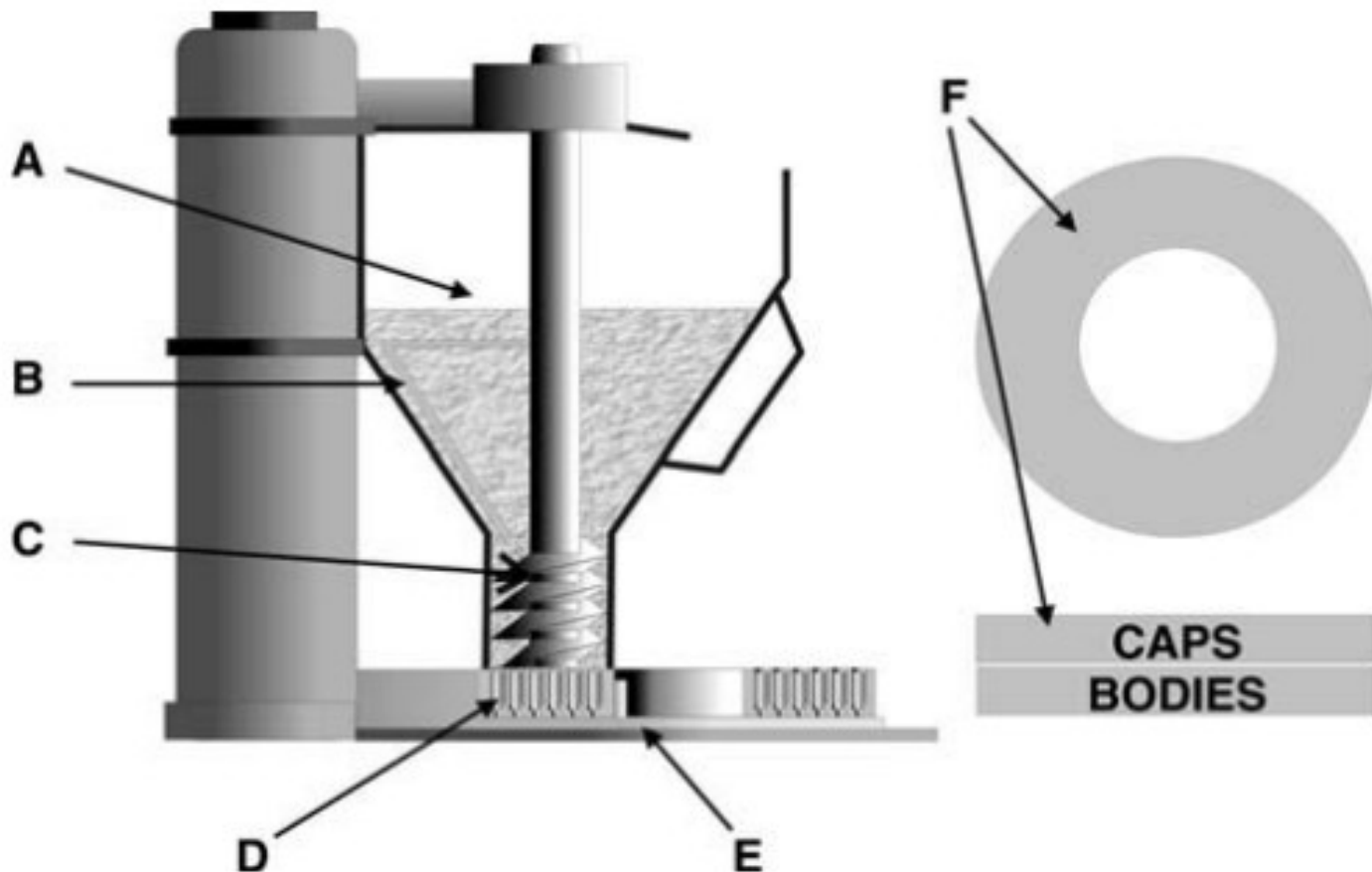
2. Fully Automatic Machines

- Capsule- filling machines may vary widely in their engineering design, the main difference between them from a formulation point of view is **the means by which the formulation is dosed into the capsules.**
- Main dosing methods may be identified for powder filling

Capsule – filling machines...

1. Auger Fill Principle

- Was the first successful industrial filling machine
- The empty capsule bodies are held in a filling ring, which rotates on a turntable under the powder hopper.
- The fill of the capsules is primarily volumetric.
- Because the auger mounted in the hopper rotates at a constant rate, the rate of delivery of powder to the capsules tends to be constant.
- Consequently, the **major control** over fill weight is the **rate of rotation** of the filling ring under the hopper

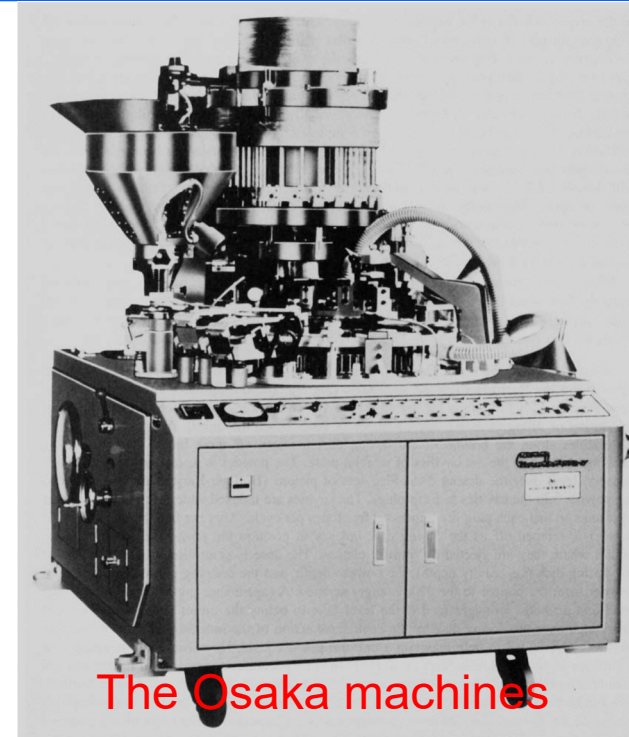


Schematic diagram of auger filling system (Model No.8): (A) powder hopper; (B) stirrer arm; (C) auger; (D) body ring holder; (E) turntable; and (F) capsule carrying rings

Capsule – filling machines...

2. Vibratory Fill Principle

- uses a vibrating plate in the powder hopper.
- The bodies in their holders passed under the hopper.
- The cavities with the bodies were overfilled and raised up against a metal plate to push the excess powder.
- This system works well with dense, free flowing material.



Capsule – filling machines...

3. Piston-Tamp Principle

- Most capsules are filled on piston-tamp machines.
- These are fully automatic fillers in which pistons or tamping pins lightly compress the individual doses of powders into plugs (sometimes referred to as “**slugs**”) and eject the plugs into the empty capsule bodies.
- The compression forces are low, about 50 100-fold less than typical tablet compression forces

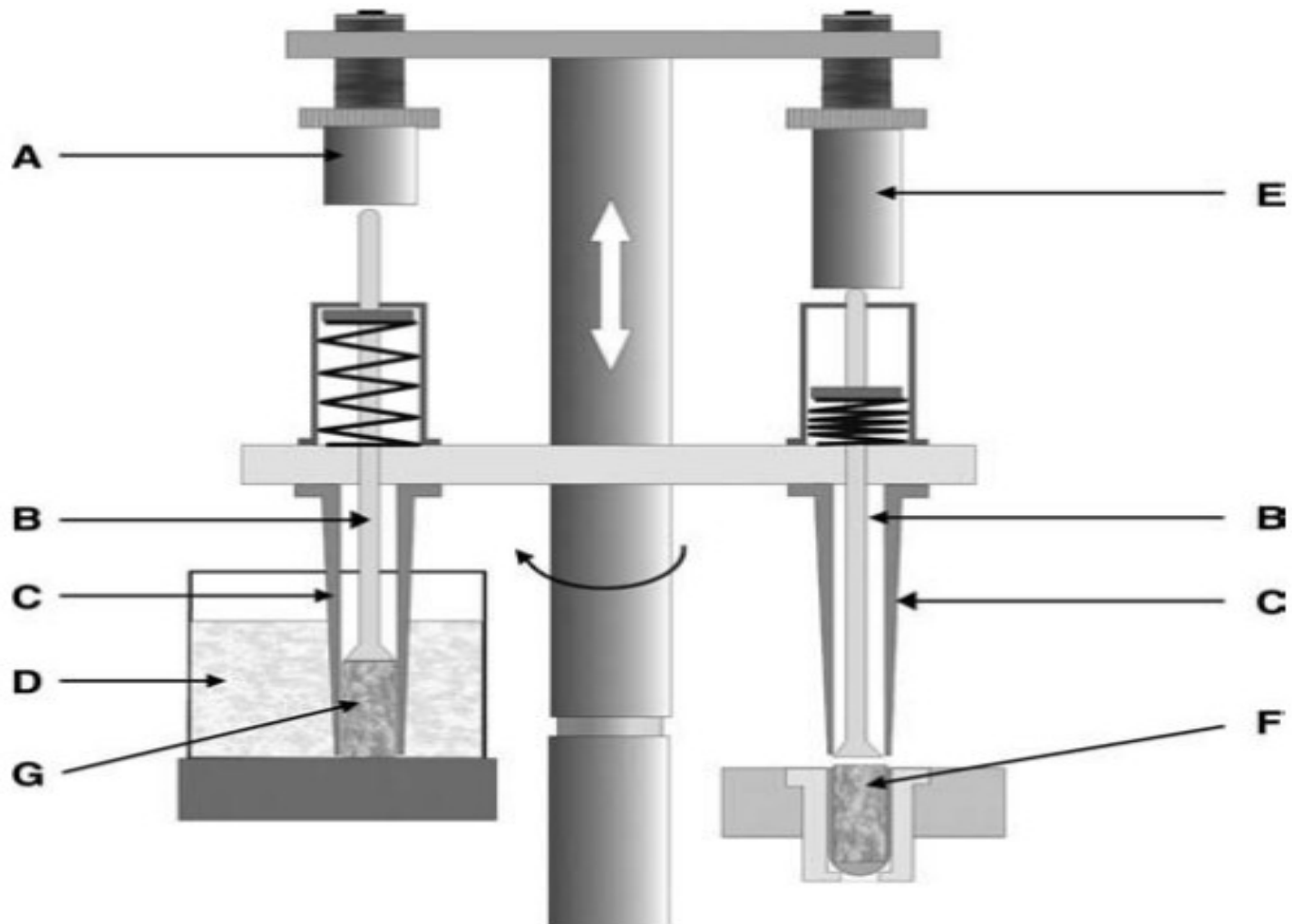
Capsule – filling machines...

3.Piston-Tamp Principle...

- There are two types of piston-tamp fillers: dosator machines and dosing-disc machines.

A. dosator machines (the dosing tube)

- ✓ tube is the most widely used, and it originated in Italy.
- ✓ The plug is formed inside a tube with a moveable piston that controls the dosing volume and applies a force to form the plug.



powder hopper; compression force platen; piston; plug
ejection platen; capsule body in bush; and powder plug;
dosing tube

Diagram of a dosator or dosing tube system (Zanasi RM63): (A) compression force platen; (B) piston; (C) dosing tube; (D) powder hopper; (E) plug ejection platen; (F) capsule body in bush; and (G) powder plug.

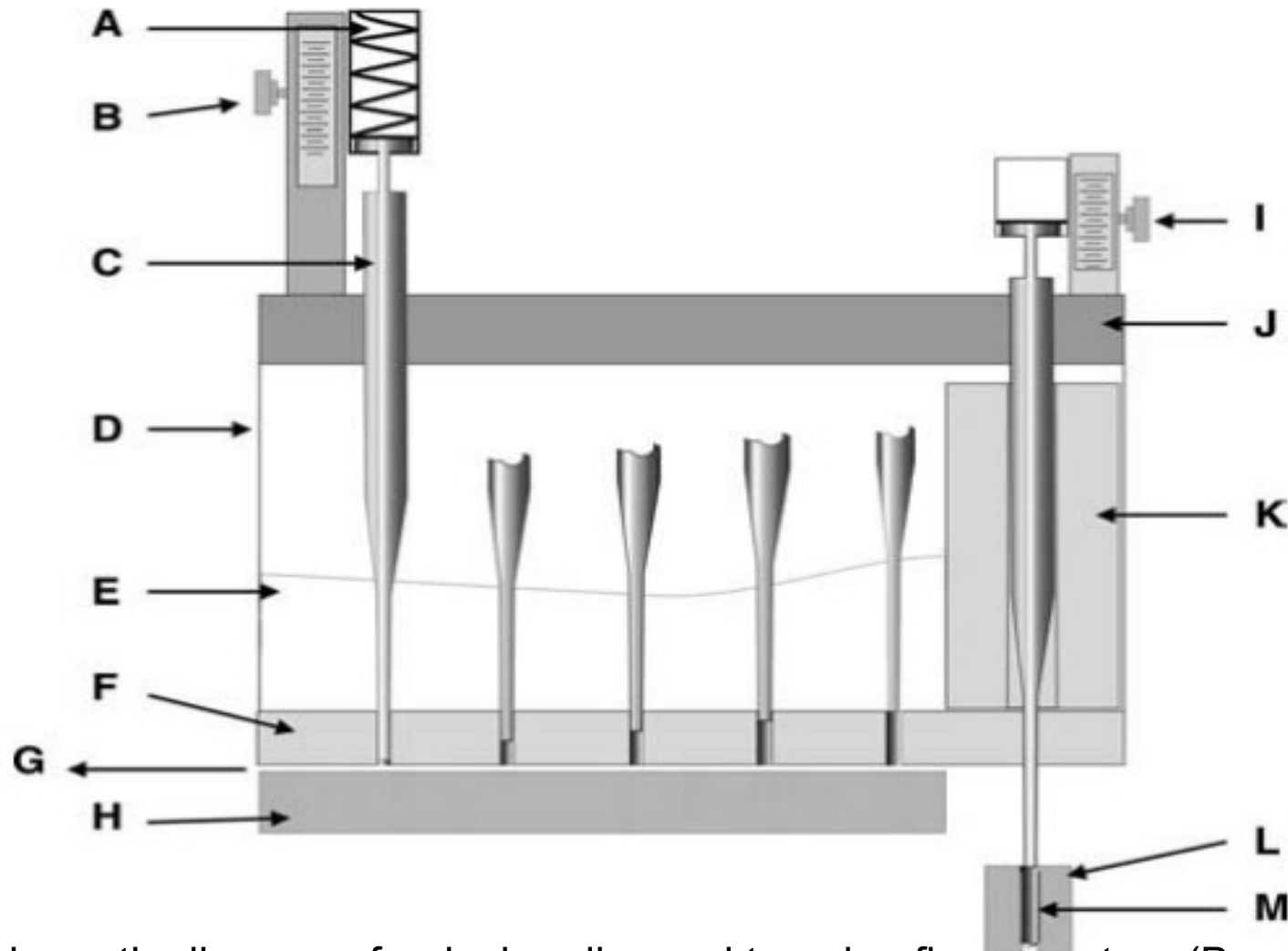
Capsule – filling machines...

B. Dosing-Disc Machines

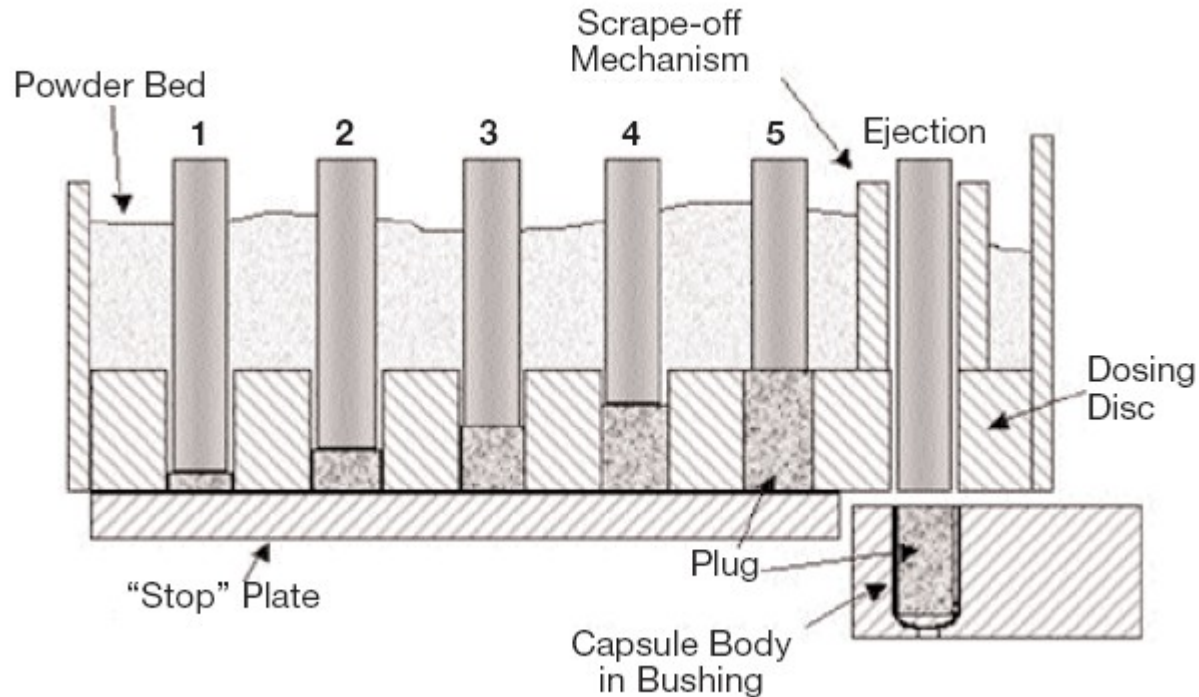
- form a plug in a similar but different manner.
- The dosing disk, which forms the base of the powder hopper, has up to six sets of machined holes.
- In a holder, above the powder hopper, there are sets of stainless steel tamping fingers corresponding to the holes in the disk.
- the tamping fingers are lowered into the powder bed.

Capsule – filling machines...

- The fingers are set to different levels, and they penetrate into the plate and consolidate the powder in the cavities into plugs.
- The dosing disks are produced in a range of thicknesses for each size of capsule.
- the selection of the correct thickness of disk for fill weight,



Schematic diagram of a dosing disc and tamping finger system (Bosch GKF machine): (A) over-load relief spring; (B) tamp-depth adjuster; (C) tamping finger; (D) powder hopper; (E) powder bed; (F) dosing disc; (G) suction; (H) support plate; (I) ejection adjuster; (J) guide block; (K) transfer block; (L) capsule body in bushing; and (M) powder plug



For a given formulation, size of tooling and powder depth over the disc, the fill weight achieved is determined primarily by the thickness of the dosing disc and the piston penetration setting (or tamping force)

Dry Solid Filling

- **Granules, pellets, and tablets** can be filled into capsules using automatic filling machines.
- Products are prepared in these forms to:
 - ✓ modify the release rates of active ingredients
 - ✓ to separate incompatible components
 - ✓ or to densify a product to achieve the fill weight in a specific size of capsule.
- The machines that can be used to fill these products can be:
 - the direct
 - indirect categories

Dry Solid Filling

- The direct
 - the material is fed into the body until it is completely full
- The indirect
 - the dose of material is measured in a chamber with a variable volume, which can be adjusted to give the required weight
 - Machines have modified dosators that either use suction to hold the material in the tube during transfer
 - or are filled when they pushed up through the material bed

Dry Solid Filling

- Properties of **tablets** are filled into capsules
 - ✓ convex and have diameters that enable them to be introduced easily into the body.
 - ✓ with sufficient clearance
- **Granules** and **pellets** should be regular in shape so that they flow and pack well.
- Their size should be related to the size of the capsule

Liquid Filling

- Two types of liquid fills
 1. nonaqueous solutions and suspensions
 2. formulations that are liquefied only for the filling process by either heat or shear stress



- ✓ will need to be sealed after filling
- ✓ The dose of material is measured using volumetric pumps.
- ✓ the uniformity of fill is in most cases better than a powder

filling machine

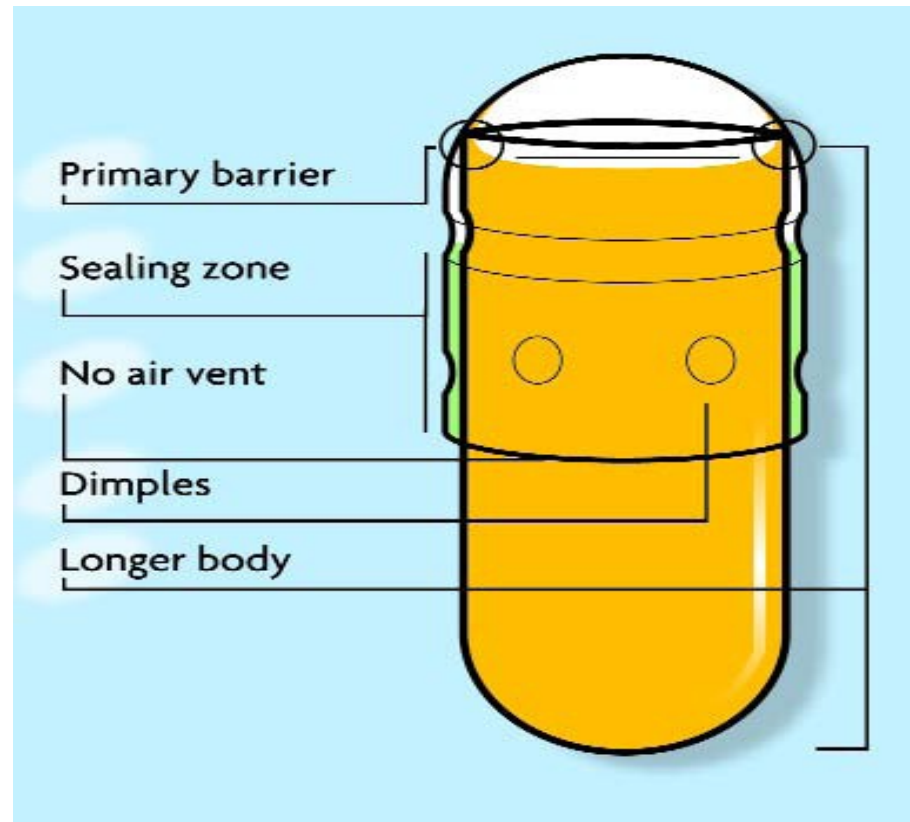
Liquid Filling

The goal of filling two-piece hard gelatin capsules with liquids is to overcome:

- poor aqueous solubility to improve oral bio-availability;
- poor content uniformity of low-dose drugs; to provide good accuracy
- and safety concerns when handling highly potent APIs, such as cytotoxins
- Low-melting point compounds are also suitable candidates for liquid filling

A capsule for liquids

- This capsule **is longer than standard capsules**, so that when the capsule body and cap are fully joined, the top of the capsule body's wall contacts the interior of the cap.
- This provides the primary barrier to prevent the liquid fill from escaping.



A capsule for liquids

- To further prevent or reduce leakage and contamination at the cap-body interface, the capsule **has no side air vents**, which are typical of capsules used in high-speed powder filling.
- The capsule is normally filled to **no more than 90 percent** of its volume to minimize the chance of the liquid fill contaminating the cap-body interface.
- This **fusion process** begins with an application of less than 50 microliters of sealing solution to the cap-body interface.
- The solution penetrates the overlapping cap and body by capillary action, while a vacuum removes any excess sealing fluid from the capsule

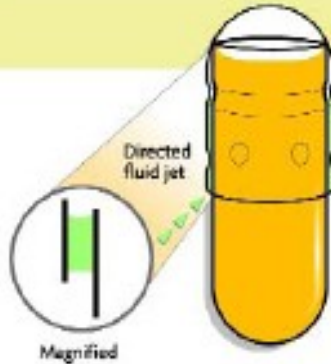
A capsule for liquids

- Next, gentle application of warm (40° to 60°C) air fuses the gelatin of the cap and body together and evaporates the sealing solution.
- The entire process takes less than 1 minute and transforms the two-piece hard capsule into a leak-free dosage unit.
- Once sealed, the capsule meets tamper-evidence guidelines since it cannot be opened without visibly altering it.

A capsule for liquids

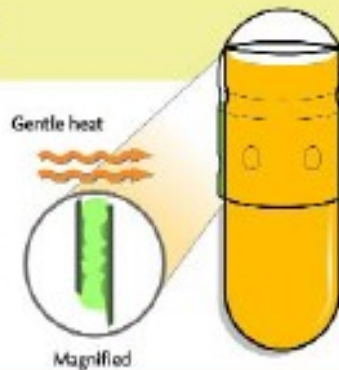
STAGE ● Spraying

Sealing fluid is sprayed onto cap-body interface. Capillary action draws fluid up between the cap and body. Vacuum removes excess fluid.



STAGE ● Warming

Applying gentle heat completes melting and fusion to form a leak-free, tamper-evident seal.



STAGE ● Setting

Capsules set and harden at room temperature.



filling machines



Sealing machines

Advantage liquid-filled hard-gelatin capsules

- Advantage liquid-filled hard-gelatin capsules have over soft-gelatins is the manufacturing process of the hard-gelatin capsules.
 - Since two-piece capsule manufacturing does not require some of the solvents commonly used in the manufacturing of soft-gelatin capsules
- Vitamins, in particular, have benefited from the reemergence of the liquid-filled hard gelatin capsules.
 - Oily products, such as vitamins E and A, can be filled and sealed in this manner without adding thickening agents.
 - This allows maximum concentration of the substance with minimum capsule size, while also reducing the risk of capsule instability.

Encapsulation of oily liquids and pastes in hard shell capsules





- There are several benefits to gelatin band sealing, including:
 - Preventing fill leakage.
 - Reducing oxygen permeability.
 - Providing a tamper evident band.
 - Ensuring capsules will not open during swallowing, preventing esophageal injuries.



Step -4: Cleaning and polishing the filled capsules

- Small amount of powder may adhere to the outside of capsules after filling.
- The powder may be bitter or otherwise unpalatable and should be removed before packaging or dispensing.
- On a small scale, capsules may be cleaned individually or in small numbers by rubbing them with a clean gauze or cloth.
- On large scale, many capsules filling machines are affixed with a cleaning vacuum that removes any extraneous material from the capsules as they exit the equipment.

Finishing

- Dedusting and/or polishing operation are completed before the remaining operations of inspection, bottling, and labeling.
- The following are the methods most commonly used
 - Pan polishing
 - Cloth dusting
 - Brushing
- **UNFILLED CAPSULE SORTING**
 - After Polishing, the capsules are feed to the 'Unfilled Capsule Sorter'.
 - Here, unfilled capsules, partly filled capsules, loose capsule bodies or loose capsule caps, if any, are sorted out. Good quality capsules are now ready for packing

Special types of hard gelatin

- **Altered Release**
- **Coating capsules**
- **Sustained release capsules**
- **Liquid filled hard gelatin capsules**
- **Rectal capsules**
- **Capsules for packing of Ophthalmic ointments**

Soft Gelatin Capsules



Description of the soft gel

- Soft gelatin capsule (soft gel) is a soft, globular, gelatin shell somewhat thicker than that of hard gelatin capsule.
- Although virtually they can be made any shape, oval or oblong shapes are usually selected for oral administration.

Gelatin shell formulation

- Typical soft gel shells are made up of **gelatin, plasticizer**, and materials that impart the desired appearance (**colourants and/or opacifiers**), and **preservatives** sometimes **flavours**.

Gelatin shell formulation

Gelatin

- Both type A and type B gelatin are used.
- It normally constitutes 40% of the wet molten gel mass.

Gelatin shell formulation.....

□ Plasticizers

- Plasticizers are used to make the soft gel shell elastic.
- They usually account for 20-30% of the wet gel formulation.
- The most common plasticizer used in soft gels is **glycerol**,
- Although **sorbitol** and **propylene glycol** are also frequently used, often in combination with glycerol.

Gelatin shell formulation.....

- The amount and choice of the plasticizer may affect the hardness of the final product and may even affect its dissolution or disintegration characteristics, as well as its physical and chemical stability.
- One of the most important aspect of softgel formulation is to ensure that there is minimum interaction or migration between the liquid fill matrix and the softgel shell.

Gelatin shell formulation.....

❑ Water

- Water usually accounts for 30-40% of the wet gel formulation.
- Its presence is important to ensure proper processing during gel preparation and softgel encapsulation.

Gelatin shell formulation.....

- Following encapsulation, excess water is removed from the softgels through controlled drying.
- In dry softgels the equilibrium water content is typically in the range 5-8% w/w, which represents the proportion of water that is bound to the gelatin in the softgel shell.

Gelatin shell formulation.....

□ Colourants and opacifiers

- Colorants (soluble dyes, or insoluble pigments) and opacifiers are typically used at low concentrations in the wet gel formulation.
- An opacifier, usually **titanium dioxide**, may be added to produce an opaque shell when the fill formulation is a suspension, or to prevent photodegradation of light-sensitive fill ingredients.

Types of softgel fill formulations

- ***Lipophilic liquids/oils***: e.g., soya bean oil
- ***Hydrophilic liquids***
 - ✓ Polar liquids with a sufficiently high molecular weight are commonly used. e.g., PEG 400.
 - ✓ Below 10% by weight of ethanol or water can be used.
- ***Suspensions***
- ***Solid powders***

Manufacture of softgels

❑ The Rotary-die process

- Rotary die encapsulation is the process by which gel ribbon and unit dose of liquid fill matrix are combined to form the softgel.
- Before the encapsulation process takes place, there are two sub-processes that are often carried out simultaneously, yielding the two components of a softgel.

Manufacture of softgels...

- These are
 - ✓ preparation of the gel mass which will provide the softgel shell, and
 - ✓ preparation of fill matrix.
- The gel mass is prepared by dissolving the gelatin in water at approximately 80°C and under vacuum, followed by the addition of the plasticizer.

Manufacture of softgels...

- Once the gelatin is fully dissolved, other components, such as colours, opacifier, flavours and preservatives, may be added.
- The hot gel mass is then supplied to the encapsulation machine through heated transfer pipes to a **casting drum** that forms two separate gelatin ribbons.

Manufacture of softgels...

- During the casting process the gelatin passes through the sol-gel transition and the thickness of each gel ribbon is controlled to ± 0.1 mm, in the range of about 0.5-1.5 mm.
- The thickness is checked regularly during the manufacturing process.

Manufacture of softgels.....

- The two gel ribbons are then carried through rollers (at which a small quantity of vegetable oil lubricant is applied) and onwards to the rotary die encapsulation tooling.
- Each ribbon provides one half of the softgel.

Manufacture of softgels.....

- It is possible to make bi-coloured softgels using gel ribbons of two different colours.
- The liquid fill matrix containing the active drug substance is manufactured separately from preparation of the molten gel.

Manufacture of softgels...

- Manufacture of the active fill formulation involves **dispersing or dissolving the drug substance in non-aqueous liquid vehicle** using conventional mixer-homogenizers.
- Accurately metered volumes of the liquid fill formulation are injected from the wedge device into the space between the gelatin ribbons as they pass between the die rolls.

Manufacture of softgels.....

- The wedge-shaped injection system is itself heated to approximately 40°C.
- The injection of liquid between the ribbons forces the gel to expand into the pockets of the dies, which govern the size and shape of the softgels.

Manufacture of softgels.....

- The ribbon continues to flow past the heated wedge injection system and is then pressed between the die rolls.
- Here the two softgel capsule halves are sealed together by the application of heat and pressure. The capsules are cut automatically from the gel ribbon by raised rims around each die on the rollers.

Manufacture of softgels.....

- During softgel encapsulation the following parameters should be controlled:
 - ✓ **Temperature.** This controls the heat available for capsule seal formation.
 - ✓ **Timing.** The timing of the dosing of unit quantities of fill formulation into the softgel during its formation.

Manufacture of softgels.....

- ✓ **Pressure.** The pressure exerted between the two rotary dies controls the **softgel shape** and the **final cut-out** from the gel ribbon.
- After manufacture, the capsules are passed through a **tumble drier** and then, to complete the drying process, spread on to **trays** and stacked in a **tunnel drier**.

Manufacture of softgels.....

- The tunnel drying process may take 2 or 3 days, or possibly as long as 2 weeks, depending on the specific softgel formulation.
- Finally, the soft gels are quality checked and packed into bulk containers in order to prevent further drying and for storage.

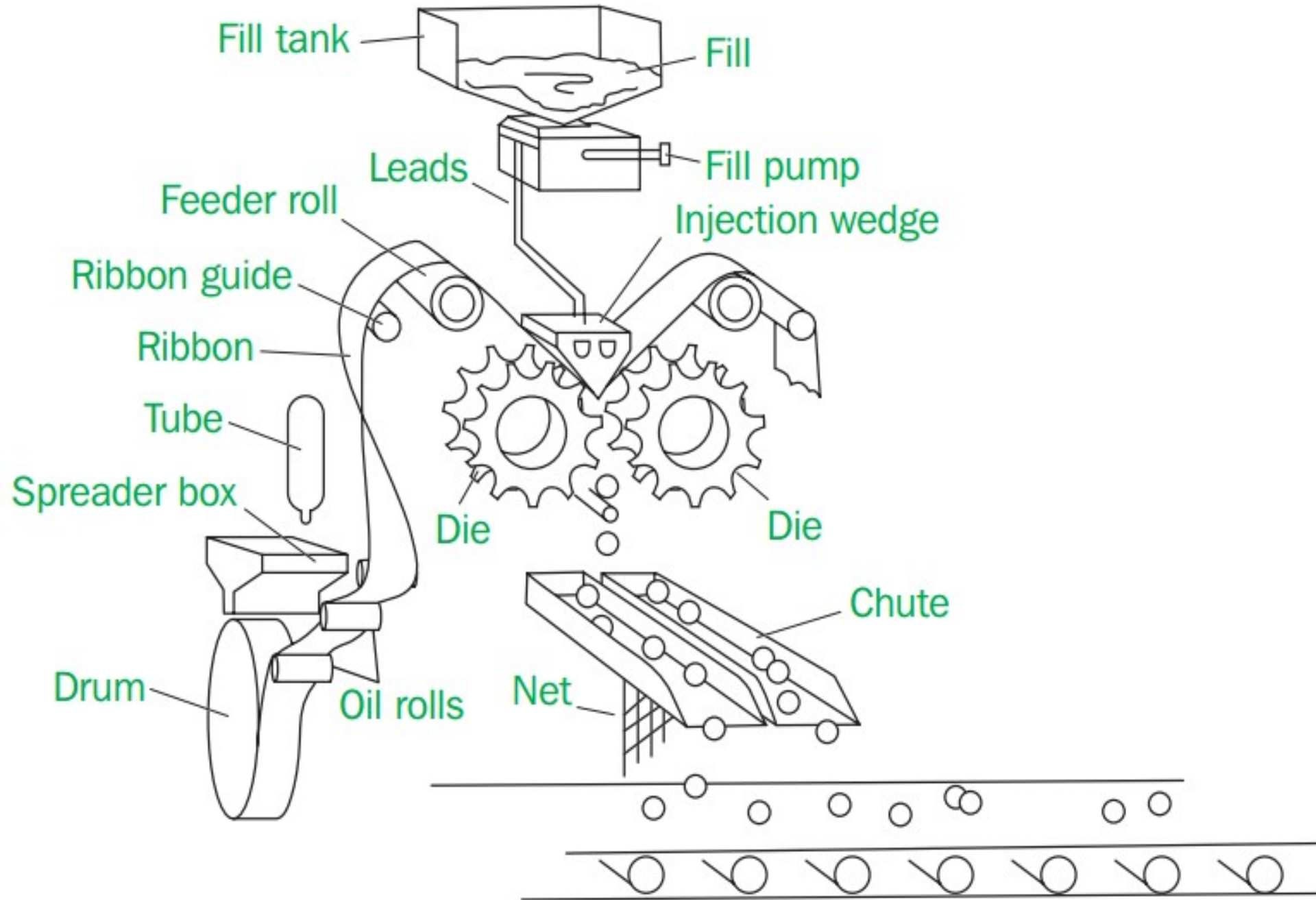
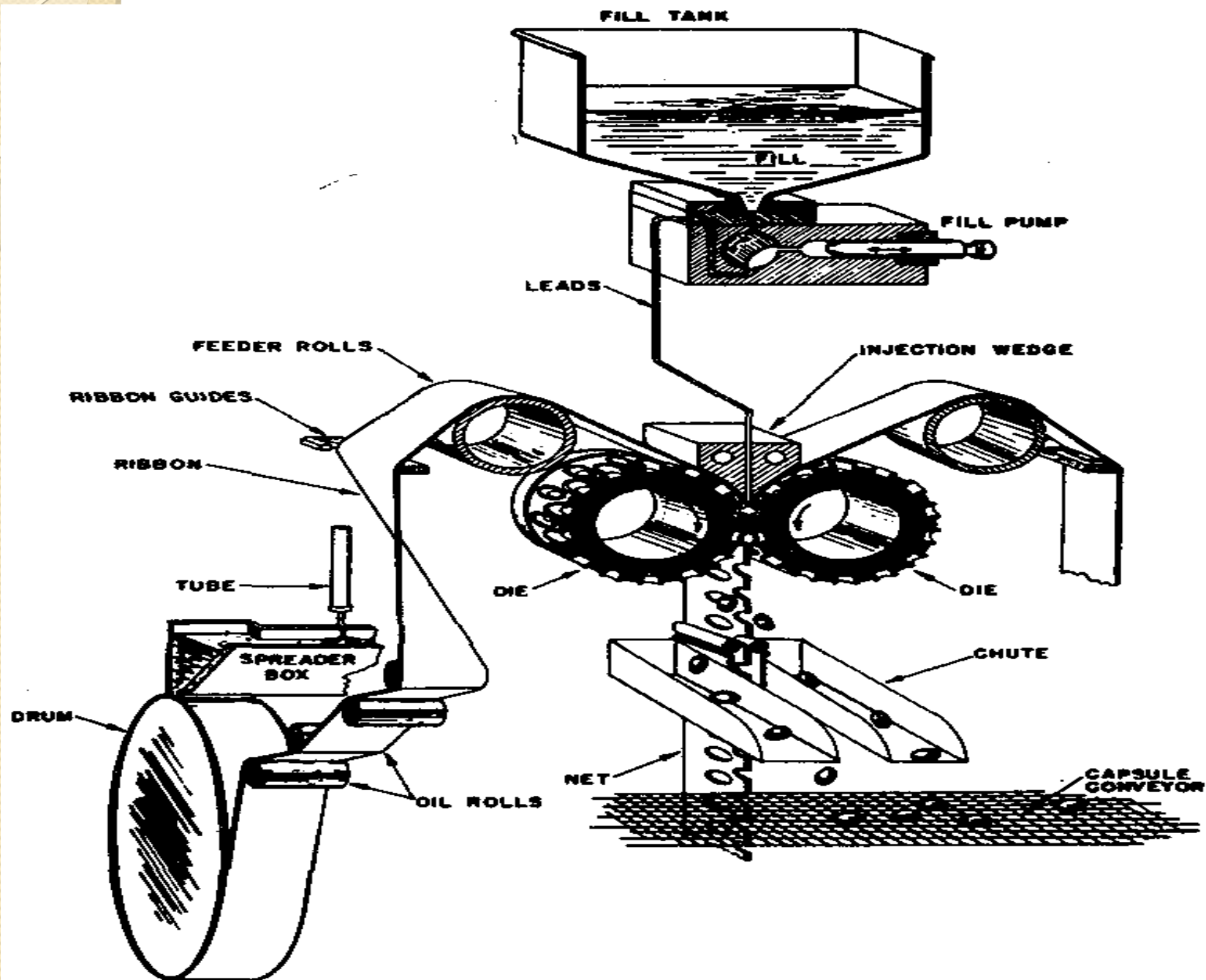


Illustration of the manufacturing method for soft gelatin capsules



Manufacture of softgels.....

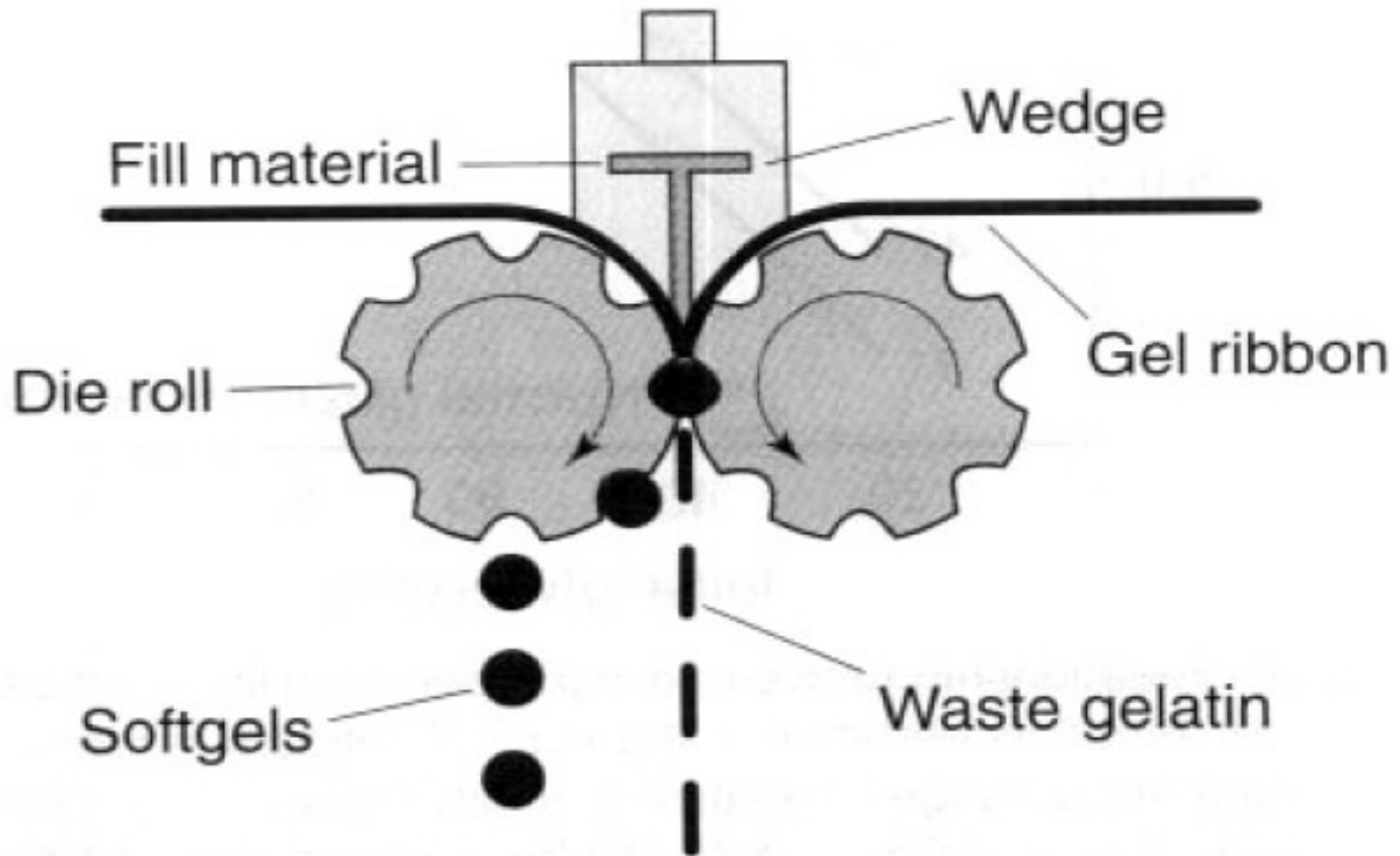


Fig. 30.8 Softgel formation mechanism.

Evaluation of capsule shells

The whole process of capsule shell manufacturing is controlled by computer and operator

Computer control

- Temperature
- Viscosity
- Drying condition

And other factors critical to process, that should be controlled are

- ✓ Film thickness
- ✓ Dimension
- ✓ Weight
- ✓ Moisture
- ✓ Brittleness

Operator may perform

- Visual finding
- Separation property (joining & separation)
- Microbiological test

Quality control

Steps in capsule manufacturing and quality control

Production step	Corresponding quality control
-----------------	-------------------------------

1 Preparation of raw material	
--------------------------------------	--

→ Gelatin	appearance, odour, colour, grain size, solubility, gelling, (bloom), viscosity, pH value, isoelectric point, bacteriology, chemical purity
→ Water	electrolyte content, pH value, bacteriology
→ Colouring agents and pigments	identity, solubility, bacteriology, chemical purity
→ Gelatin solution	viscosity, temperature, colour shade, colour composition

Quality control

Steps in capsule manufacturing and quality control

Production step	Corresponding quality control
-----------------	-------------------------------

2 Production machine	
-----------------------------	--

→ 1 Dipping, 2 Rotation, 3 Gelling, 4 Drying, 5 Stripping, 6 Cutting, 7 Joining, 8 Ejection	temperature, relative humidity, viscosity, dimensions, colour shade, laboratory check
--	--

3 Inspection	defects
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4 Imprinting	laboratory checks on ink, statistical control
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5 Counting, packaging	final release by Quality Control
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6 Batch release	certificate
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Quality control (filled HGCs and SGC)

Reading assignment

- ☐ **Weight variation test**
- ☐ **Content uniformity**
- ☐ **Disintegration test**
- ☐ **Dissolution test,...**

LIQUID PHARMACEUTICAL PREPARATIONS

- Liquid oral preparations are composed of many types of formulations, both aqueous and nonaqueous, including solutions, suspensions, and emulsions.
- Oral solutions are homogeneous mixtures of one or more solutes dissolved in a suitable solvent or mixture of mutually miscible solvents.
- In pharmaceutical terms, solutions are defined as “liquid preparations that contain one or more soluble chemical substances, usually dissolved in water.
- Solutions are classified on the basis of physical properties, method of preparation, use, and type of ingredients.



- A disperse system is defined as a heterogenous, two- phase system in which the internal (dispersed, dis- continuous) phase is distributed or dispersed within the continuous (external) phase or vehicle.
- Molecular dispersions are homogeneous in character and form true solutions.
- Colloidal dispersions are intermediate in size between true solutions and coarse dispersions.

Classification Scheme of Disperse Systems on the Basis of the Particle Size of the Dispersed Phase

Category	Range of particle size	Characteristics of system	Examples
Molecular dispersion	$< 1.0 \text{ nm}$	Particles invisible by electron microscopy; pass through semipermeable membranes; generally rapid diffusion	Oxygen molecules, potassium and chloride ions dissolved in water
Colloidal dispersion	$1.0 \text{ nm} - 1.0 \mu\text{m}$	Particles not resolved by ordinary microscope but visible by electron microscopy; pass through filter paper but not semipermeable membranes; generally slow diffusion	Colloidal silver sols, surfactant micelles in an aqueous phase, aqueous latices and pseudolatices
Coarse dispersion	$> 1.0 \mu\text{m}$	Particles visible by ordinary microscopy; do not pass through normal filter paper or semipermeable membranes	Pharmaceutical emulsions and suspensions

APPLICATIONS

- **Patient compliance**
- easily **adjusted** as fractional doses by dilution to meet the needs of the patients
- Some **deliquescent and hygroscopic powders** are more easily dispensed as liquids
- Some drugs that are **not tolerated** in a concentrated form may **be less irritating** if dissolved in soothing liquid.
- Higher rate of absorption

LIMITATIONS

- Are **less stable**
- Special techniques are required to solubilize poorly soluble drugs
- **Masking the taste of inherently very bitter drugs is sometimes difficult.**
- Extremely potent drugs with **a low therapeutic index** cannot be given in an oral liquid dosage form
- Cannot be administered to **the unconscious patient.**

Formulation consideration

- The formulation of solutions presents many technical problems to the industrial pharmacist.
 - Some drugs are inherently unstable. This property is magnified when the drug is in solution
 - Special techniques are required to solubilize poorly soluble drugs
 - The final preparation must satisfy the requirements of pharmaceutical elegance with regard to taste, appearance and viscosity.

- To solve these and many other formulation problems encountered with pharmaceutical liquids, an interesting dichotomy of investigative skills is required.
- Successful formulation requires a blend of scientific acuity and pharmaceutical art.

Manufacturing consideration

1. Raw Materials

- Raw materials used in manufacturing liquids should conform to specification. These specifications should assure identity, purity, uniformity and freedom from excessive microbial contamination.
- Incoming raw materials should be improved and thoroughly tested before they are released for manufacturing.
- Additional processing may be a desirable property such as particle size or freedom from MOs.
- Typical MOs affecting drug microbiological stability are *Pseudomonas*, *Escherichia coli*, *Salmonella* , and *Staphylococcus*.
- With regard to microbial contamination of raw materials it is usually much easier to begin with low counts in the raw materials.

- Chemical instability reactions appear with or without microbiological contribution through reactions such as hydrolysis, oxidation, isomerization, and epimerization.
- Interactions between ingredients and ingredients with container closure materials are established as the principal causes of these reactions, for instance,
 - the hydrolysis of cefotaxime sodium,
 - the oxidation of vitamin C,
 - the isomerization of epinephrine,

- Aside from active ingredients, water is usually the most important constituent in a liquid product.
- Water used should meet USP requirements for purified water.
- Water may be obtained by distillation, ion exchange method, reverse osmosis, UV sterilization, and Membrane filtration.

COMPONENTS AND ADDITIVES

- **Sweetening Agents**

- They are used **to mask bitter or unacceptable tastes** of constituents.
- Sweetening agents constitute a major portion of the solid content in most liquid oral dosage forms.
- The most commonly used sweeteners include **sucrose, sorbitol, mannitol, liquid glucose, honey molasses, saccharin, and aspartame.**

- Official Simple Syrup is an 85% w/v solution of sucrose in water.
- It is frequently used in conjunction with **sorbitol, glycerin, and other polyols**, which reduce its tendency to crystallize.

- **Flavoring Agents**
- **Coloring Agents**
- **Preservatives--** Liquid oral preparations are the most likely of all nonsterile pharmaceutical products to be contaminated by micro-organisms.

- **Antioxidants**--Many drugs in solution are subject to oxidative degradation.
- ✓ **Sulfites** are the most common antioxidants in **aqueous** solutions.
- ✓ Certain compounds (**e.g., ascorbic and citric acids**) have been found to act as synergists, increasing the effectiveness of antioxidants, particularly those that block oxidative reactions.
- Frequently, **chelating agents such as EDTA** are used in formulations containing trace amounts of heavy metals that would otherwise catalyze oxidative reactions.

Viscosity-Controlling Agents

- It is sometimes desirable to increase the viscosity of a liquid to provide or to improve **palatability or pourability**.
- This can be achieved by viscosity-controlling agents such as **PVP** or **CMC**. These compounds give aqueous solutions that are stable over a wide pH range.

2. Equipments

- Equipments used in the manufacture of oral solutions consists
 - ✓ Mixing tanks equipped with a means of agitation
 - ✓ Measuring devices for large and small amount of solids and liquids
 - ✓ Filtration system for the final polishing, sterilization of the solution.
 - ✓ System for bulk material handling, discharging

All equipments must be thoroughly cleaned and sanitized before use

Manufacturing solutions

- For both small-and large-scale manufacture of solutions the only equipment necessary is suitable **mixing vessel, a means of agitation** and **a filtration system** to ensure clarity of the final solution.
- During manufacture, the solute is simply added to the solvent in a mixing vessel and stirring is continued until dissolution is complete.

Manufacturing solutions....

- If the solute is more soluble at elevated temperatures it may be advantageous to apply heat to the vessel.
- Care must be taken, however, should any volatile or thermolabile materials be present.

Manufacturing solutions....

- Size reduction of solid materials to increase their surface areas should also speed up the process of dissolution.
- Solutes present in low concentrations, particularly dyes, are often pre-dissolved in a small volume of the solvent and then added to the bulk.

Manufacturing solutions.....

- Volatile materials such as flavours where possible, added at the end of a process and after any cooling, to reduce loss by evaporation.
- Finally, it must be ensured that significant amounts of any of the materials are not irreversibly adsorbed on to the filtration medium used for final clarification.

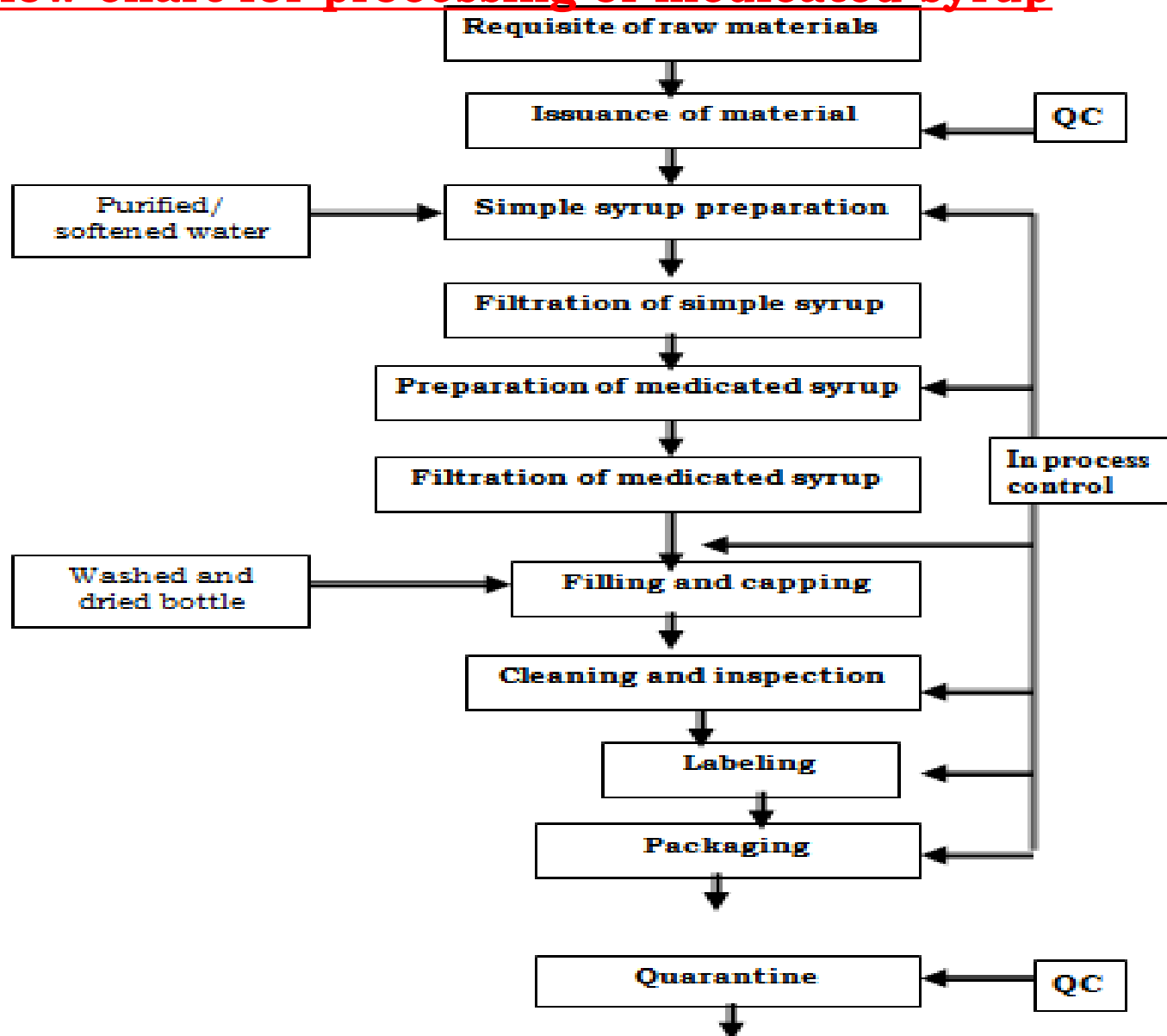


Turbine mixer

3. Compounding procedure

- Dilute solution of rapidly dissolving materials are prepared by charging the solute to the solvent and agitating until the solution is homogeneous.
- When more concentrated solutions are being made or when the solute is slowly dissolving, it is advantageous to employ heat.

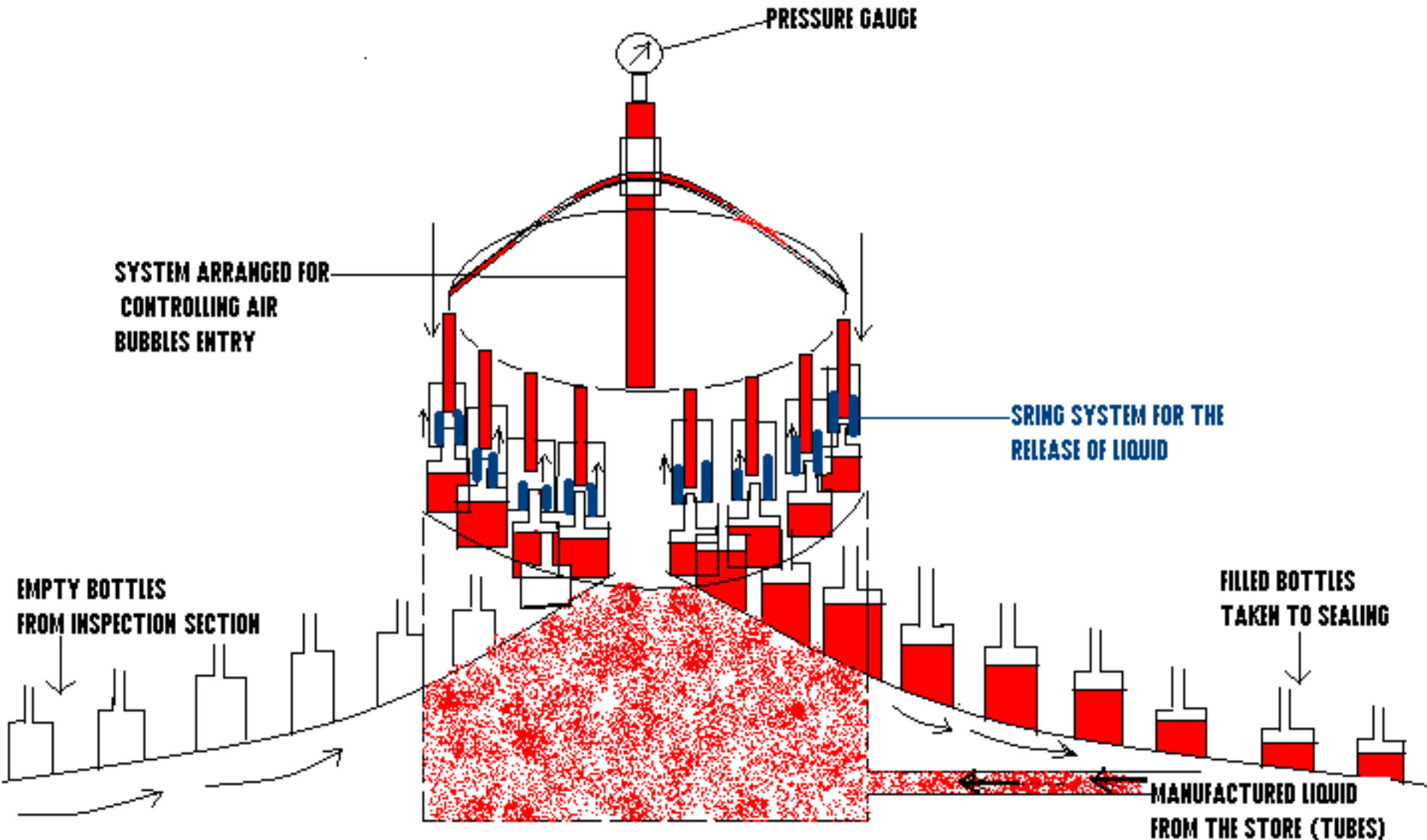
Flow chart for processing of medicated syrup



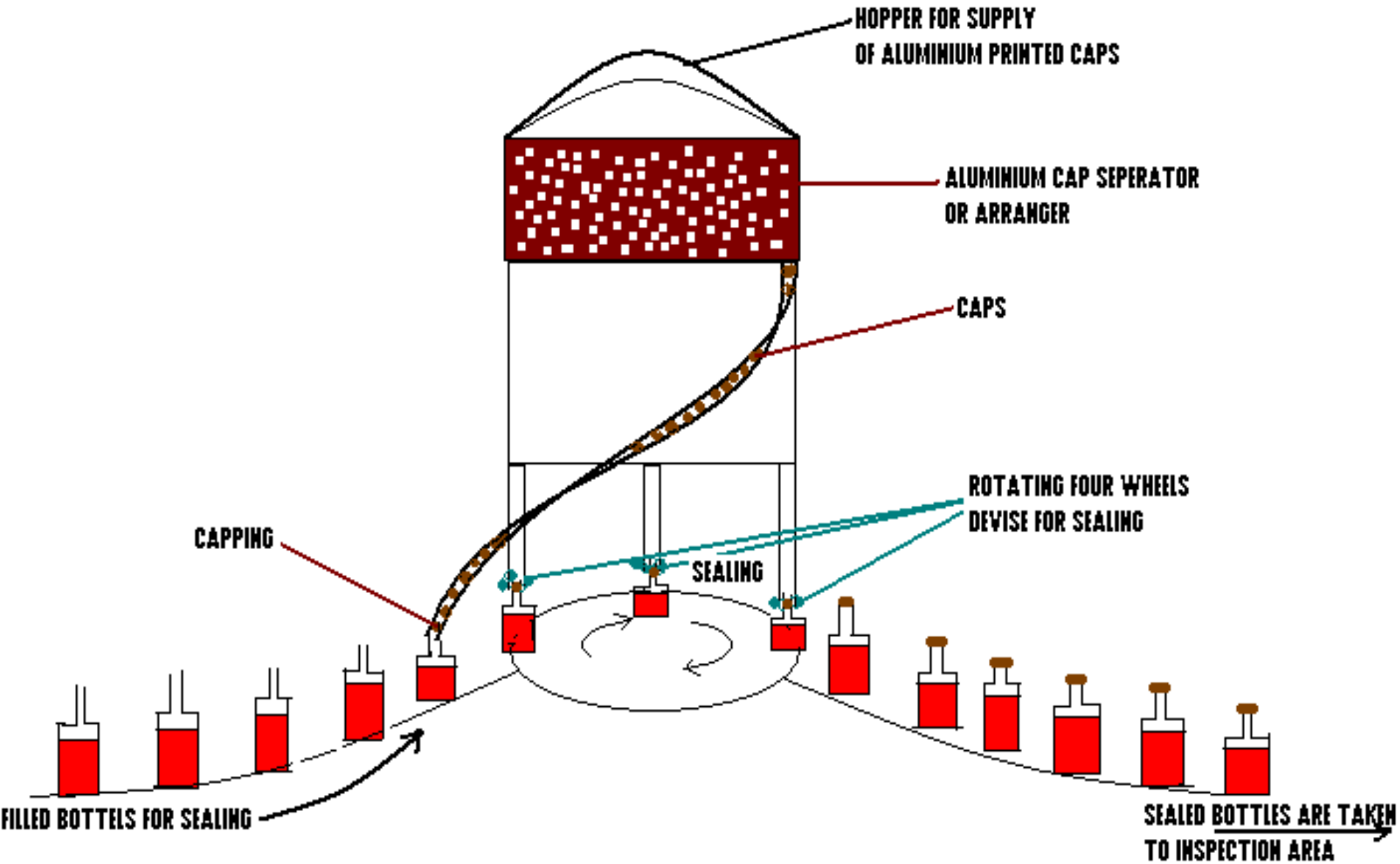
S.S. FILLING MACHINE & RITA FILLING MACHINE

– MAX 125 /min

– MAX 325/min



CAPPING AND SEALING AUTOMAT MACHINE



•Capping and sealing is done in the same machine.

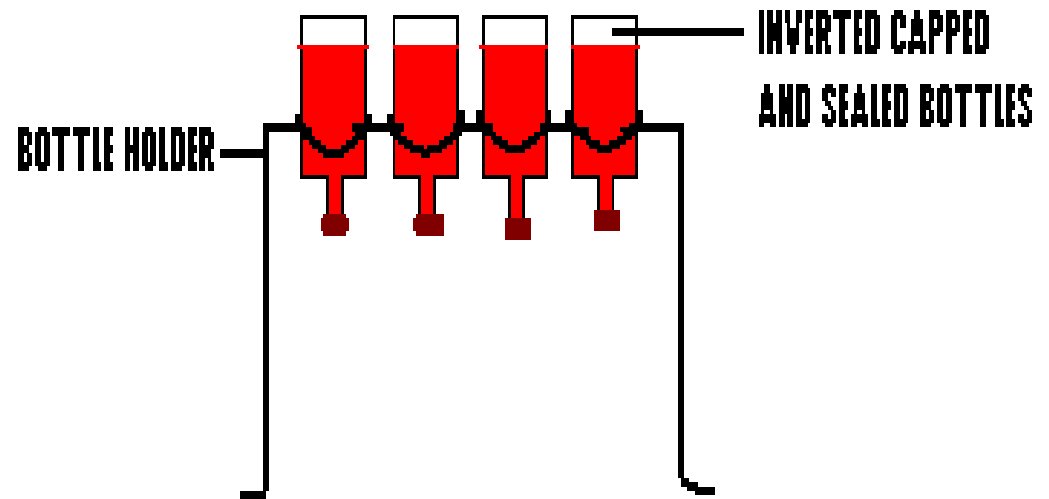
Inspection:-

1. Whether the bottles are capped properly are examined.

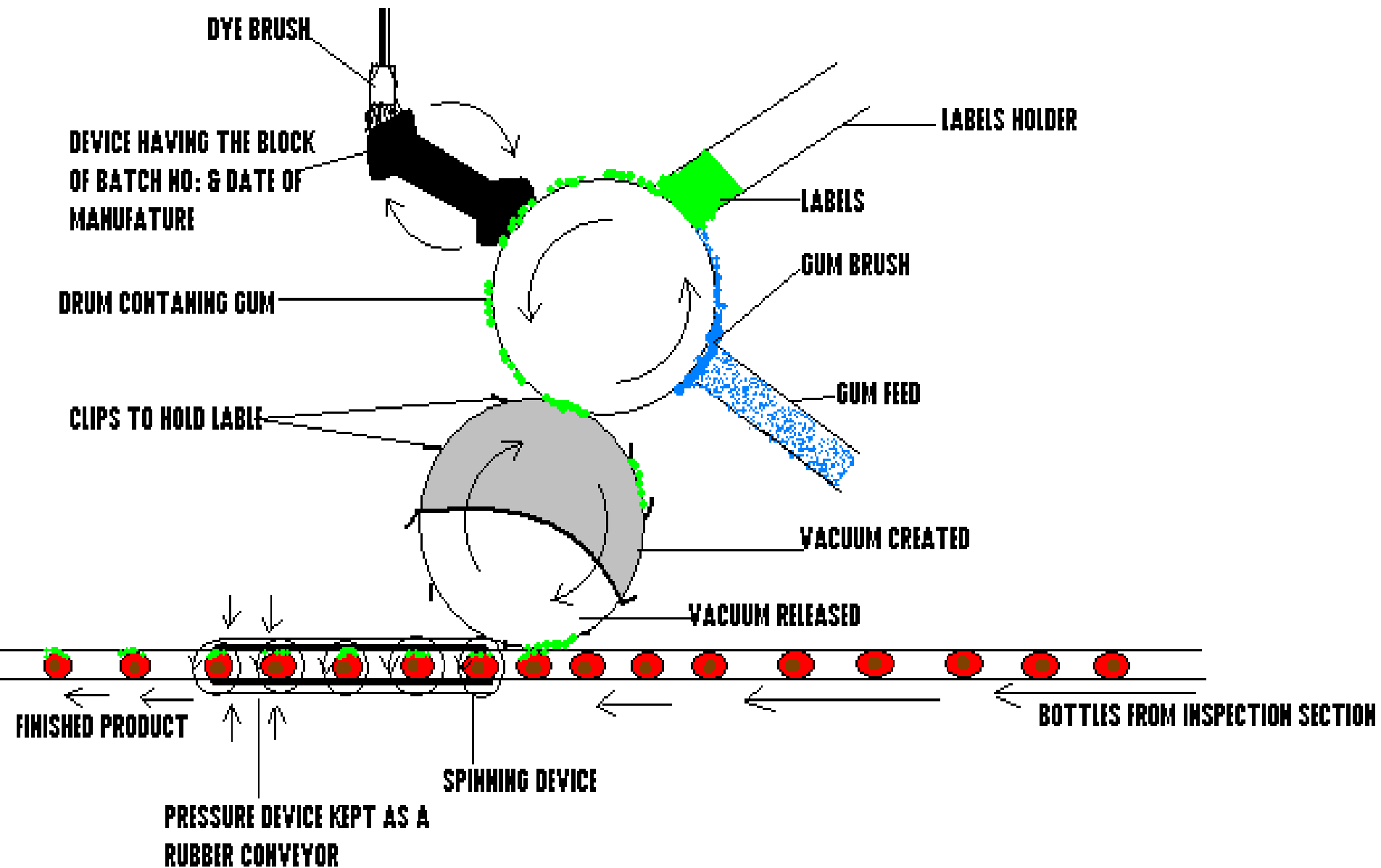
2. Whether the bottles have any leakage of liquid is examined.

* Leakage is observed by means of keeping the bottles inverted.

* If there is any leakage through the bottle cap, then they are rejected.



Labeling

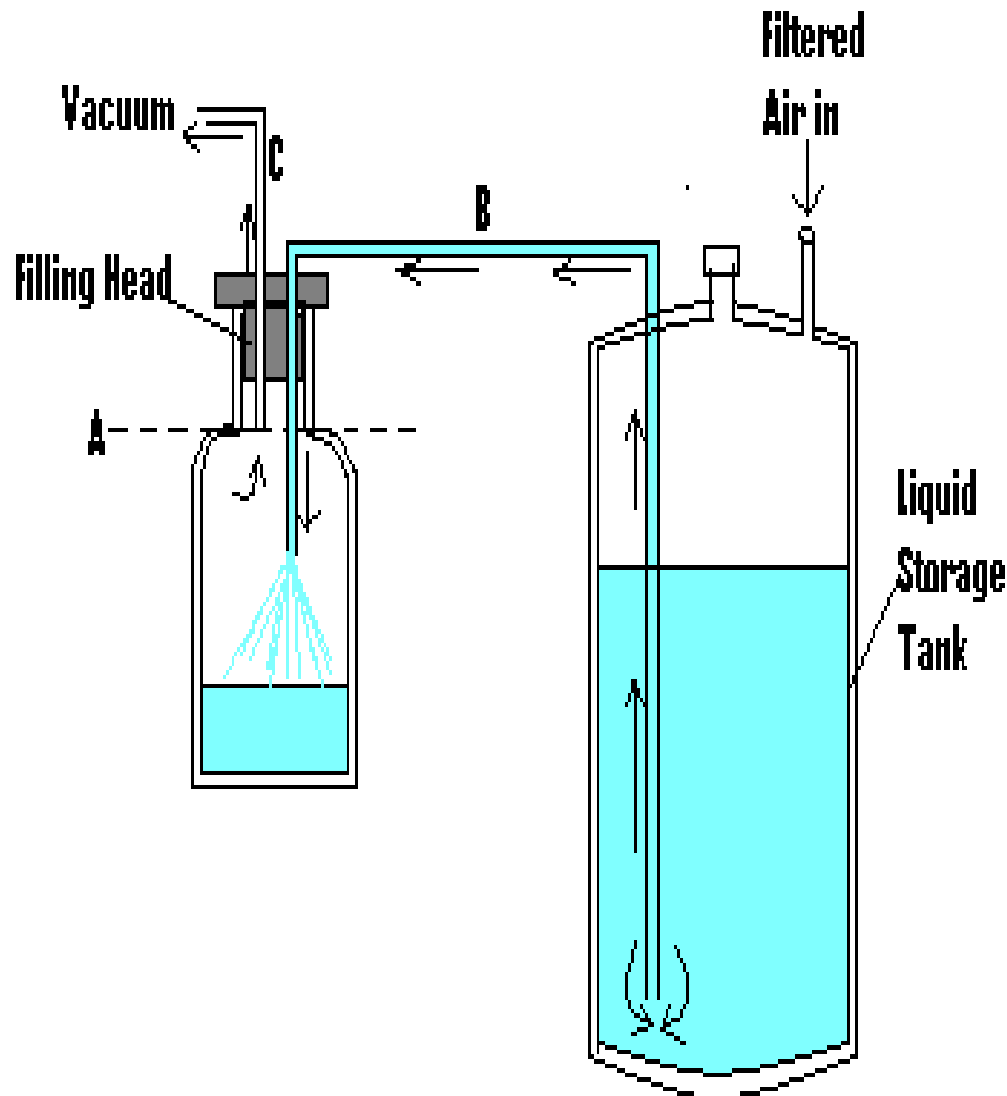


Filling and Packaging

- Methods used for filling a pharmaceutical liquid varies depending on the characteristics of liquid (e.g.: viscosity), types of package into which the liquid is placed and the required production output.

The principle of vacuum filling

Schematic diagrams of the principle used in vacuum filling.



The vacuum drawn through tube C reduced pressure in the container. Pressure differential causes liquid in storage tank to flow through tube B into container. When liquid level reaches point A, excess is carried out through tube C and may be recycled to liquid storage tank.

Note-- In gravity- vacuum filling, the the bottom of the liquid storage tank is at a level above the filling head. In pressure-vacuum filling, a positive pressure applied to the liquid in the storage tank.

CRITICAL ASPECTS OF LIQUIDS MANUFACTURING PROCESS

a. Physical Plant

Heating, Ventilation, and Air Conditioning System/HVAC

- ✓ For heat - labile drugs, uncontrolled temperature increments can activate autooxidation chains when the drug product ingredients react with oxygen and generate free radicals but without drastic external interference.
- ✓ Vitamins, essential oils, and almost all fats and oils can be oxidized.
- ✓ Clindamycin, which has to be stored at room temperature and away from excess heat and moisture
- ✓ The tocopherols, some esters of gallic acid, as well as BHA and BHT (butylated hydroxyanisole and butylated hydroxytoluene) are common antioxidants used in the pharmaceutical industry

-  **Isolation of Processes**--- To minimize cross - contamination and microbiological contamination.
 - For instance, steroids and sulfas require more isolation than over - the - counter (OTC) oral products
 - To minimize exposure of personnel to drug aerosols and loss of product, a sealed pressure vessel must be used to compound aerosol suspensions and emulsions

Dust Removal System

Air Filtration System

b. Equipment

c. Particle Size of Raw Materials

Formula	Per ml	Per batch (5000 L)
Drug	2.00mg.	10.0kg.
Sodium Benzoate USP	1.00mg.	5.0kg.
Menthol, USP	0.10mg.	0.5kg.
Alcohol, USP	0.05ml. (40.8mg.)	250.0liters (204.0 kg.)
Flavor	0.005ml. (4.5 mg.)	25.0 liters (22.5 kg.)
Dye FD &C Yellow No:6	0.10mg	0.5kg.
Glycerin	0.05ml. (62.45 mg.)	250.0 liters (312.250kg)
Sorbitol solution, USP	0.10 ml. (128.5mg)	500.0 liters (642.5kg.)
Standard granulated sugar	550.00mg.	2750.0kg.
Purified water ,USP q.s to	1.0ml.	5000 liters

Production of Suspension

Definition

- A Pharmaceutical suspension is a coarse dispersion in which internal phase is dispersed uniformly throughout the external phase.
- The internal phase consisting of insoluble solid particles having a specific range of size which is maintained uniformly through out the suspending vehicle with aid of single or combination of suspending agent.

Classification Scheme of Disperse Systems on the Basis of the Physical State of the Dispersed Phase and the Dispersion Medium

Dispersed phase	Dispersion medium		
	Solid	Liquid	Gas
Solid	Solid suspension (zinc oxide paste, toothpaste)	Suspension (tetracycline oral suspension USP, bentonite magma NF)	Solid aerosol (epinephrine bitartrate inhalation aerosol USP, smoke)
Liquid	Solid emulsion (hydrophilic petrolatum USP, butter)	Emulsion (mineral oil emulsion USP, Milk)	Liquid aerosol (nasal sprays, fog)
Gas	Solid foam Foamed plastics Pumice	Foam Effervescent salts in water Carbonated beverages	None

- The external phase (suspending medium) is generally aqueous in some instance, may be an organic or oily liquid for non oral use.

Classification

Based on General Classes

- Oral suspension
- Externally applied suspension
- Parenteral suspension

Formulation Components

Components	Function
API	Active drug substances
Wetting agents	They are added to disperse solids in continuous liquid phase.
Flocculating agents	They are added to floc the drug particles
Thickeners	They are added to increase the viscosity of suspension.
Buffers and pH adjusting agents	They are added to stabilize the suspension to a desired pH range.
Osmotic agents	They are added to adjust osmotic pressure comparable to biological fluid.
Coloring agents	They are added to impart desired color to suspension and improve elegance.
Preservatives	They are added to prevent microbial growth.
External liquid vehicle	They are added to construct structure of the final suspension.

- **Packaging:** Using various type of glass and plastics

Preparing technique

- The actual preparation of suspension involves choosing the ingredients and determining the type of manufacturing equipment to be used.
- Suspensions can be prepared by
 - Dispersion method
 - Precipitation method
 - Dry powder to be reconstituted to suspension

Cotrimoxazole Oral Suspension

Bill of Materials			
Scale (mg/mL)	Item	Material Name	Quantity/L (g)
40.00	1	Trimethoprim micronized (98% particles below 50 microns)	8.00
200.00	2	Sulfamethoxazole powder (100% particles below 50 microns)	40.00
20.00	3	Magnesium aluminum silicate (Veegum HV)	4.00
22.50	4	Carboxymethylcellulose sodium	4.50
350.00	5	Glycerin	70.00
400.00	6	Propylene glycol	80.00
5.00	7	Polyvinyl pyrrolidone (polyvinyl pyrrolidone K-30)	1.00
20.00	8	Polysorbate 80	4.00
12.50	9	Colloidal silicon dioxide (aerosil 200)	2.50
375.00	10	Sorbitol (70% solution)	75.00
5.00	11	Saccharin sodium	1.00
3.00	12	Citric acid	0.60
2200.00	13	Sucrose	440.00
5.00	14	Methyl paraben	1.00
1.50	15	Propyl paraben	0.30
0.035	16	Raspberry red color	0.007
0.025	17	FD&C red no. 40	0.005
5.00	18	Banana flavor	1.00
5.00	19	Apricot flavor	1.00
—	20	Water, purified	q.s. to 1 L

Manufacturing suspensions

- First a concentrated dispersion of the suspending agent is made.
- This is best accomplished by adding the material slowly to the vehicle while mixing.
- Suitable mixers include either an impeller type of blender or a turbine mixer.
- This stage is important, as it is necessary to ensure that agglomerates of the suspending agent are fully broken up.

Manufacturing suspensions.....

- If they are not, then the surface of each agglomerate may gel and cause the powder inside to remain non-wetted.
- Very intense shearing, however, can destroy the polymeric structure of the suspending agent, and it may be better to use milder shearing and then allow the dispersion to stand until full hydration has been achieved.
- hydration may be instantaneous or may, as with tragacanth, take several hours.

Manufacturing suspensions...

- If the suspending agent is blended with water-soluble ingredients, such as sucrose, this will also aid dispersion.
- The drug to be suspended is then added in the same way, along with the wetting agent.
- For very hydrophobic drugs, wetting may be facilitated by mixing under reduced pressure.

Manufacturing suspensions...

- Other ingredients should now be added, preferably dissolved in a portion of the vehicle, and the whole made up to volume if necessary.
- Finally, homogenization would ensure complete dispersion of the drug and the production of a smooth and elegant preparation.

Production of Emulsions

Definition

- Emulsion: is liquid dosage form in which one phase is dispersed in another phase.
- Oil phase disperse in aqueous phase (o/w) or aqueous phase disperse in oil phase (w/o).
- Emulsion is mixture of two immiscible liquids that exhibits an acceptable shelf life.
- When both immiscible liquids are mechanically agitated both phase initially tend to form droplet, when the agitation is stopped, the two liquids separate.
- The life time of the droplets is increased if an emulsifier is added to the two immiscible liquids.

An emulsion has two phases:

- **Internal (disperse or discontinuous) phase**
 - Phase of emulsion which persists in droplet form
- **External (continuous) phase**
 - Phase which surround internal phase

Manufacturing emulsion

- The spontaneous formation of an emulsion is a relatively rare occurrence.
- Instead, emulsion preparation by the commonly employed dispersion method requires a sequence of processes for breaking up the internal phase into droplets and for stabilizing them in the external phase.

Manufacturing emulsion...

- Almost all methods used for breaking up the internal phase into droplets depend on force and require some sort of agitation.
- Once the initial breakup into droplets has occurred, the droplets continue to be subject to additional forces due to turbulence, which cause deformation of the droplet and further breakdown into smaller droplets.

Mechanical equipment for emulsification...

- Various types of equipment are available to effect droplet breakup and emulsification either in laboratory or in industry.
- Regardless of size and minor variations, such equipment include:
 - ✓ **Mechanical stirrers**
 - ✓ **Ultrasonifiers**
 - ✓ **Homogenizers**
 - ✓ **Colloid mills**

Mechanical equipment for emulsification...

□ Mechanical stirrers

- An emulsion may be stirred by means of various impellers mounted on shafts, which are placed directly into the system to be emulsified.
- If the viscosity of the emulsion is low, simple top-entering propeller mixers are adequate for production purposes.



❑ **Ultrasonifiers**

- The use of ultrasonic energy to produce pharmaceutical emulsion has been demonstrated and mainly laboratory size models are available
- They are expensive instrument.
- They are useful for the laboratory preparation of fluid dispersion of moderate viscosity and one extremely low particle size. Pressure 150 to 350ps.

Mechanical equipment for emulsification...

□ Homogenizers

- Impeller types of equipment frequently produce a satisfactory emulsion; however, for further reduction in particle size (up to 1 nm), homogenizers are employed.
- In a homogenizer, the dispersion of two liquids is achieved by forcing their mixture through a small inlet orifice at high pressures.

Mechanical equipment for emulsification...

- Homogenizers may be used in one of two ways:
 - ✓ the ingredients of the emulsion are mixed and then passed through the homogenizer to produce the final product.
 - ✓ a coarse emulsion is prepared in some other way and then passed through a homogenizer for the purpose of decreasing the particle size and obtaining a greater degree of uniformity and stability.



❑ Colloid mills

- Homogenizers and ultrasonic equipment depend on sudden change in pressure to effect the dispersion of liquids; colloid mills operate on the principle of high shear, which is normally generated between the rotor and the stator of the mill.
- The principle of operation of the colloid mill is the passage of the mixed phases of an emulsion formula between a stator and a high-speed rotor revolving at speeds of 2,000 to 18,000 rpm.

Mechanical equipment for emulsification...

- The clearance between the rotor and stator is very small, but adjustable from 0.001 inches and up.
- The emulsion mixture, in passing between the rotor and stator, is subjected to a tremendous shearing action that affects a fine dispersion of uniform size.

Mechanical equipment for emulsification...

- The shearing forces applied in the colloid mill usually result in a temperature increase within the emulsion.
- It may be necessary, therefore, to use jacketed equipment to cool the emulsion during processing.

Quality control

☐ Solutions

- Assay, fill volume, clarity, pH (for weakly acidic or basic drugs)

☐ Suspensions

- Assay, fill volume, appearance, re-suspendability, viscosity, particle size and size distribution

☐ Emulsions

- Assay, fill volume, appearance, viscosity, particle size and size distribution.